

✅ This Document Step by Step procedure to set up the following:

1. Python Environment Setup,
2. Visual Studio Code Setup,
3. Anaconda Setup,
4. GitHub Account Setup,
5. Linking GitHub and VS code set up,
6. How to push the lab python notebook from VS code to GitHub,
7. How to get pull request link from GitHub and paste link in the student portal for submitting your lab work

— Step-by-Step (Beginner Friendly)

🧠 **OVERVIEW (What you're doing)**

You are:

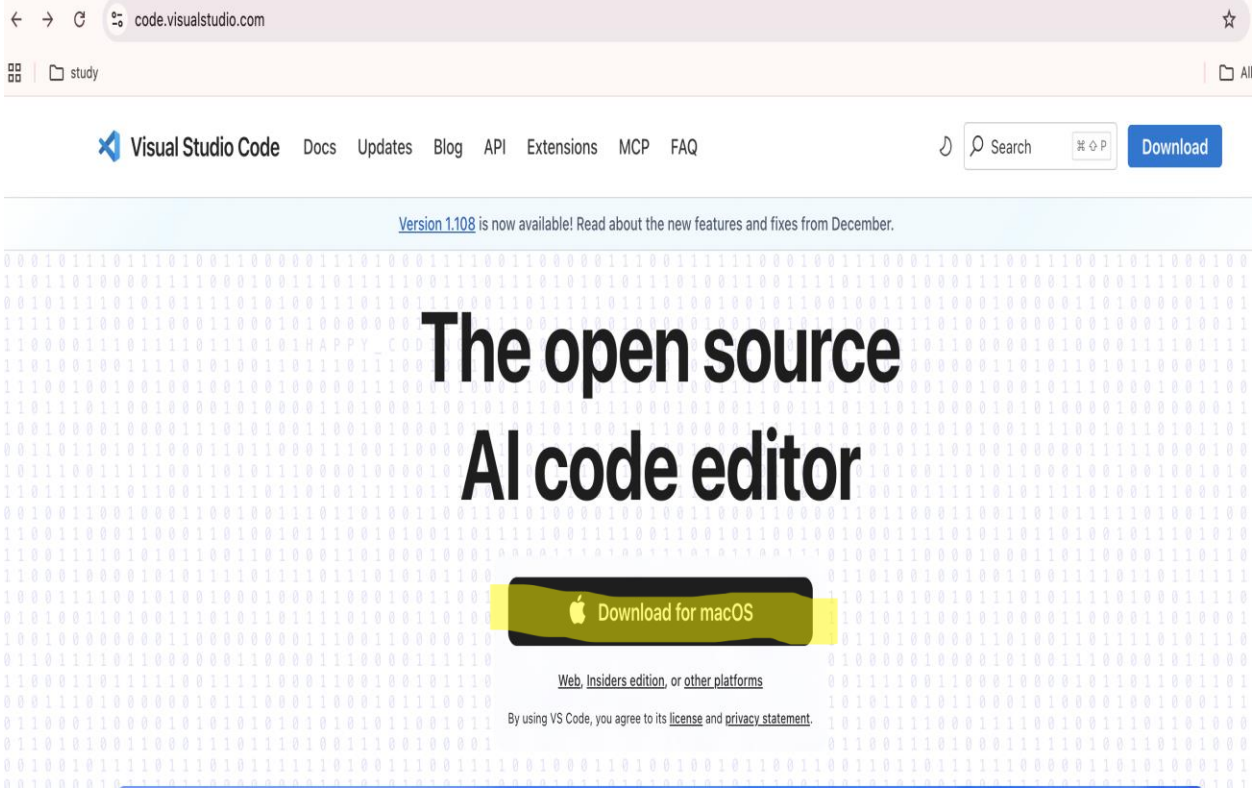
1. Installing your tools (VSCode, GitHub, Conda)
2. Connecting them together
3. Creating a safe Python “workspace” (Conda environment)
4. Running Python in a Jupyter Notebook
5. Saving the file
6. Push the file in Github
7. Copy the link and submit the link in the student portal

STEP 1 — Install VSCode (Your Code Editor)

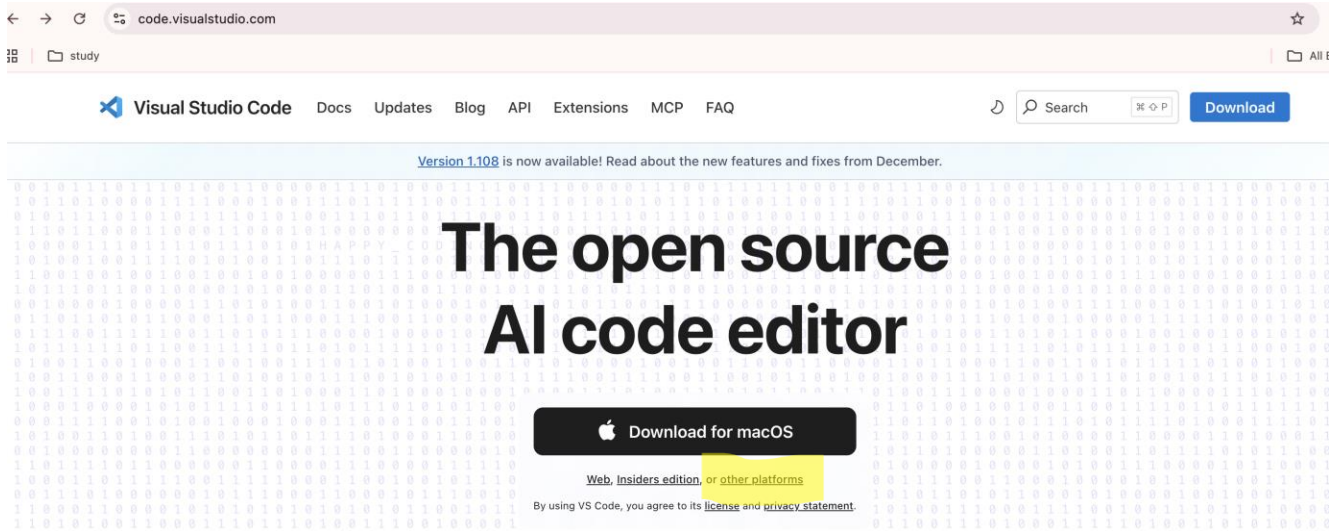
1. Download

- Go to 🖱️ <https://code.visualstudio.com>
- Download for your OS (Windows / Mac / Linux)

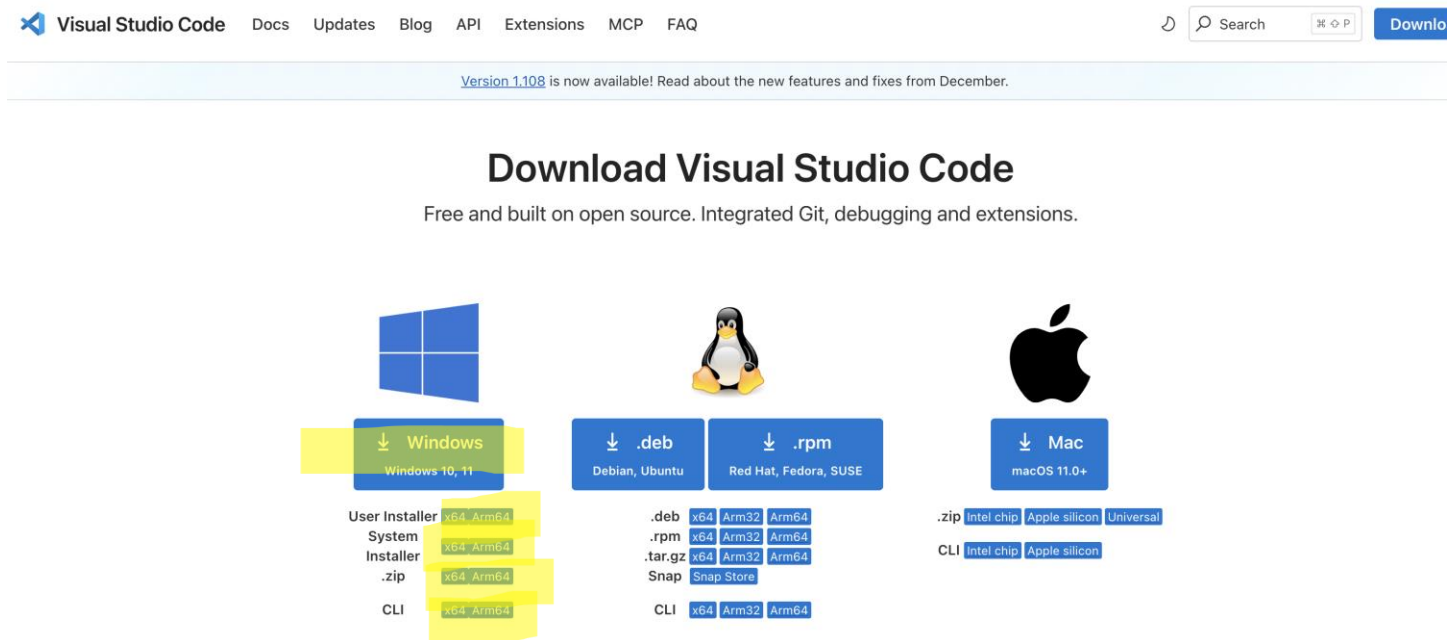
For - MacOS below is the how you download:



For - Windows below is the how you download:



You can choose the windows download options as per your laptop configuration for windows 10 & windows 11 :



2. Install

- Run the installer
- Accept defaults
- Finish installation

1) Download VS Code

1. Open your web browser.
2. Go to the official download page: <https://code.visualstudio.com/download>
3. Click the download button for your operating system (Windows, macOS, or Linux).
 - a. For Windows, this will download a .exe installer.

- b. For macOS download a .zip or .dmg.
- c. For Linux you can choose .deb, .rpm, or snap.

2) Install on Windows

1. **Run the downloaded installer** (VSCodeSetup.exe).
2. Accept the license agreement and click **Next**.
3. Choose install location (or leave default) → **Next**.
4. Choose additional tasks:
 - a. Create a desktop icon
 - b. Add “Open with Code” to context menu
 - c. Add to PATH (recommended)
→ then **Next** and **Install**.
5. When complete, click **Finish** to launch VS Code.

💡 Installing adds the code command to your system PATH so you can open folders from the terminal.

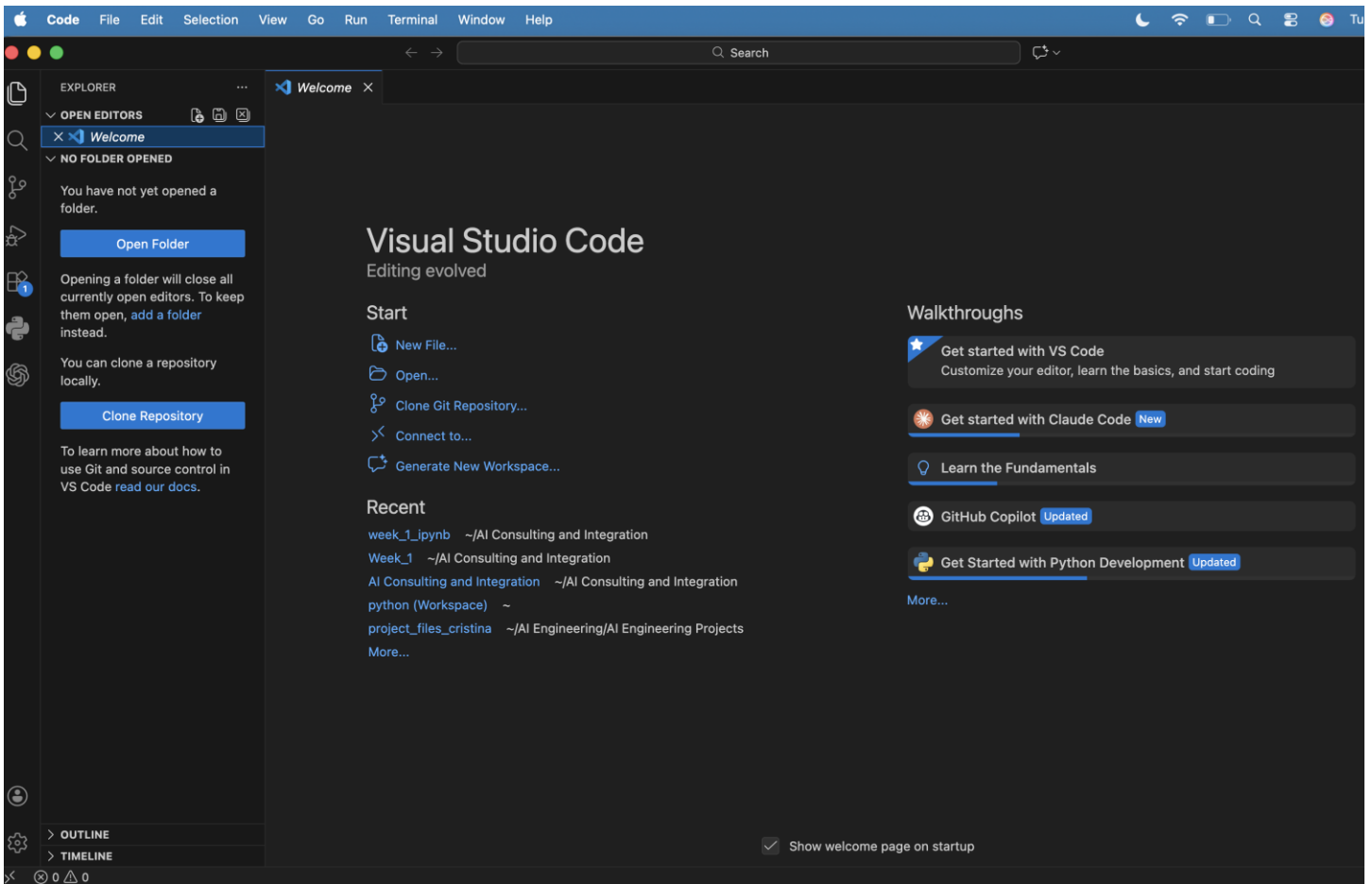
3) Install on macOS (Mac)

1. Open the downloaded .zip or .dmg.
2. Drag “Visual Studio Code.app” into your **Applications** folder.
3. Launch it from Launchpad or Finder.
4. (Optional) From VS Code: open Command Palette → “Shell Command: Install ‘code’ command in PATH” so code works in Terminal.

3. Verify

- Open VSCode
- You should see a **Welcome screen**
 - ✅ Done

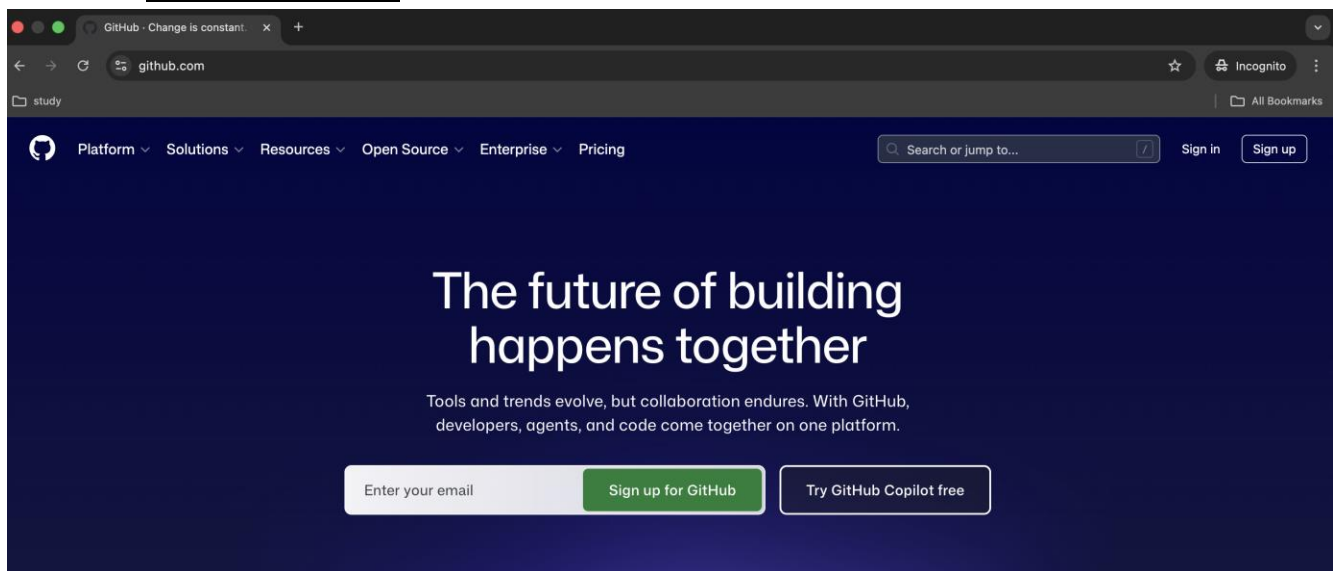
This is How your Home page of VSCode look like when you open the VSCode



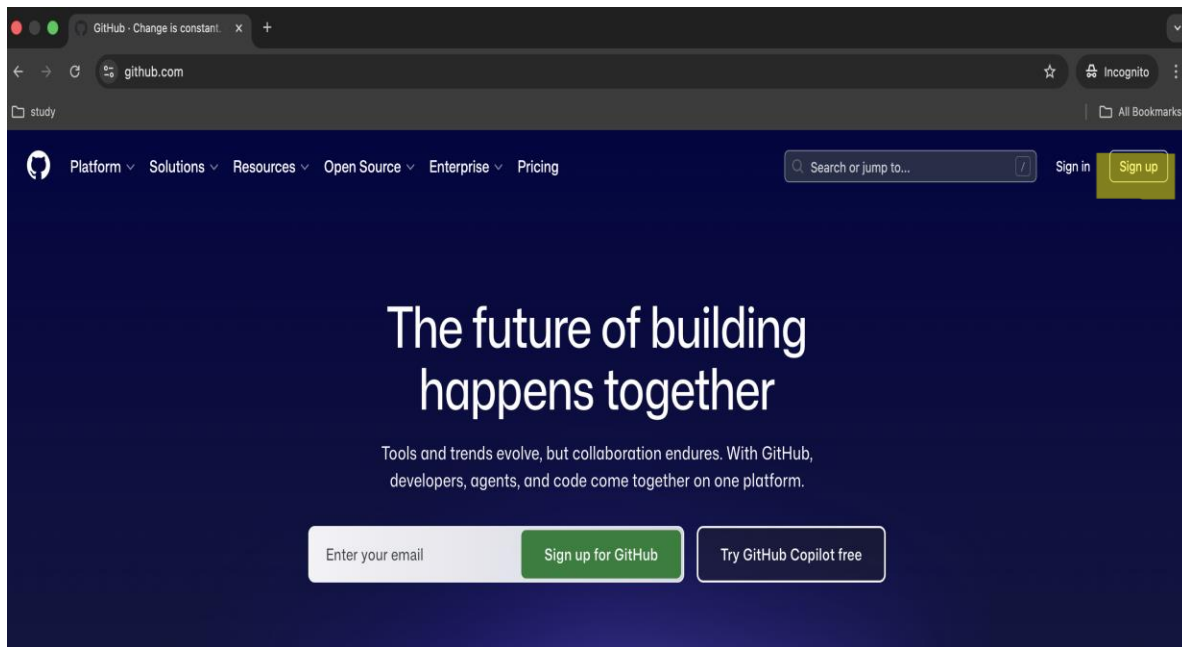
STEP 2 — Create a GitHub Account

1. Sign up

- Go to <https://github.com>



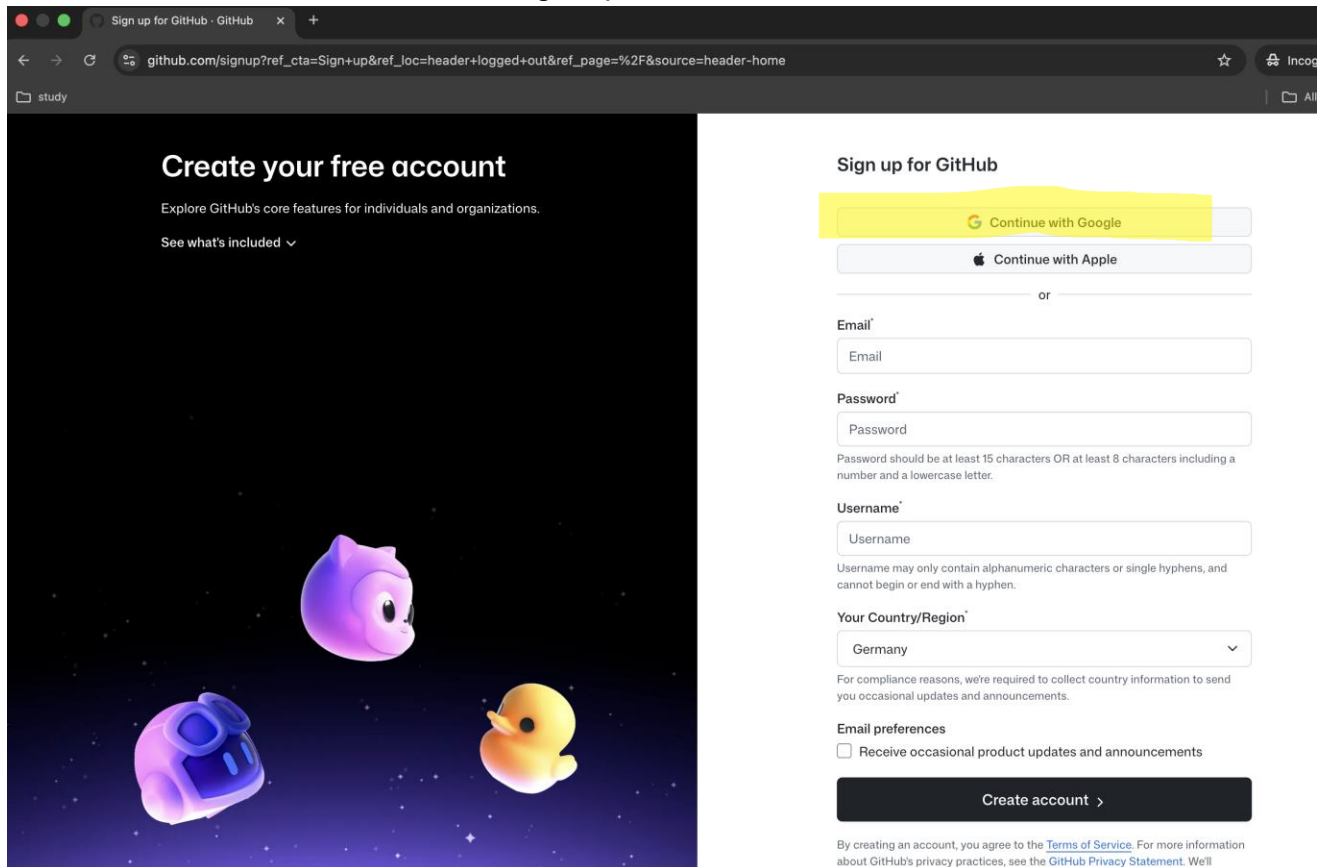
- Click **Sign up**

A screenshot of the GitHub sign-up page. The left side features the heading 'Create your free account' and a background image of three GitHub Octocat characters. The right side is a form titled 'Sign up for GitHub'. It includes options to 'Continue with Google' or 'Continue with Apple', followed by an 'or' separator. The form has input fields for 'Email', 'Password', and 'Username'. Below the 'Username' field is a note: 'Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.' There is a 'Your Country/Region' dropdown menu set to 'Germany'. Below that is an 'Email preferences' section with a checkbox for 'Receive occasional product updates and announcements'. At the bottom is a 'Create account' button. A footer note states: 'By creating an account, you agree to the Terms of Service. For more information about GitHub's privacy practices, see the GitHub Privacy Statement. We'll'.

- Give a valid email id, and password to set up your GitHub account
- Give a valid username for your GitHub account
- Choose a **professional username**
 - ❌ coolguy123
 - ✅ john-doe, jdoe-ai

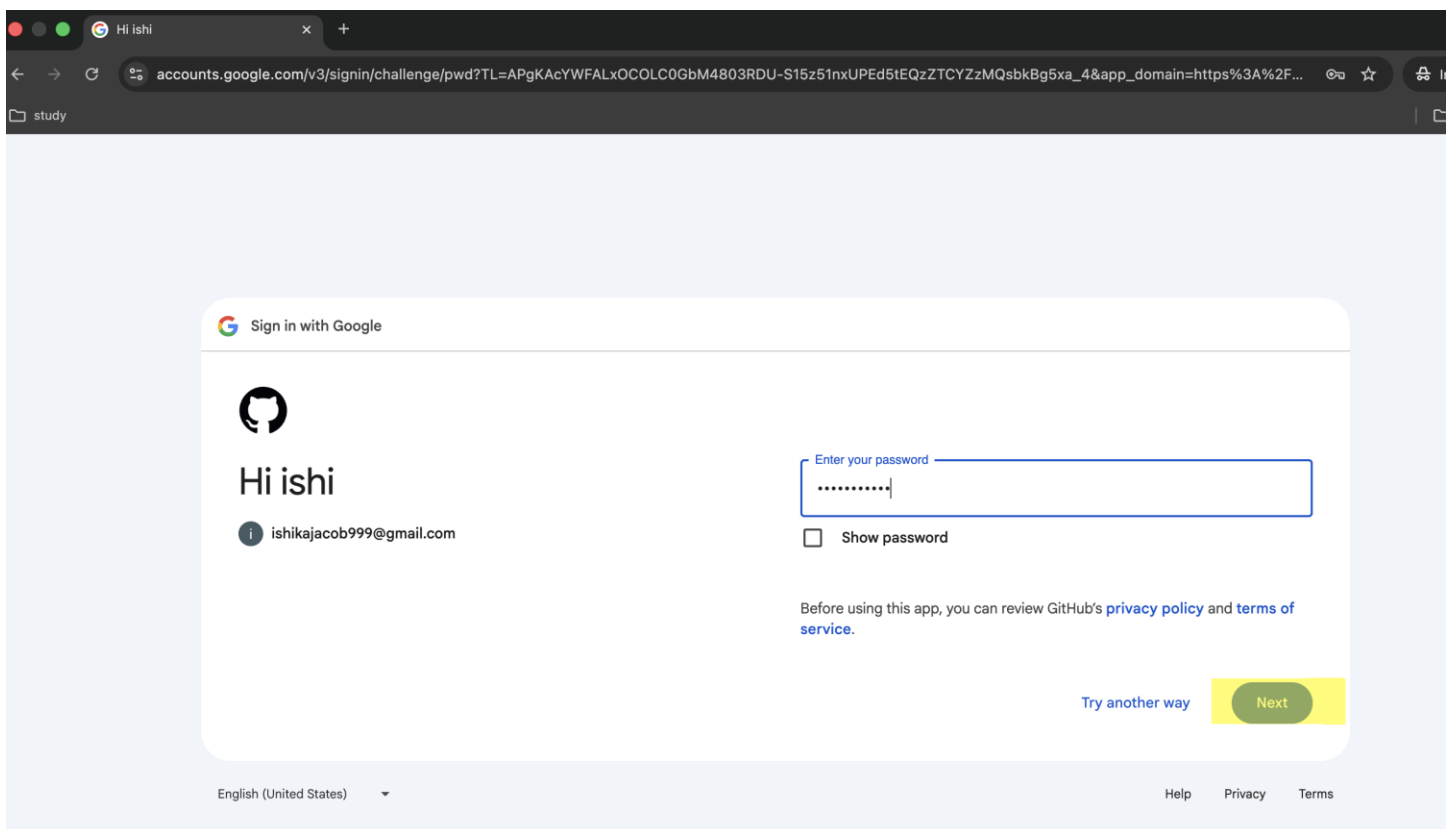
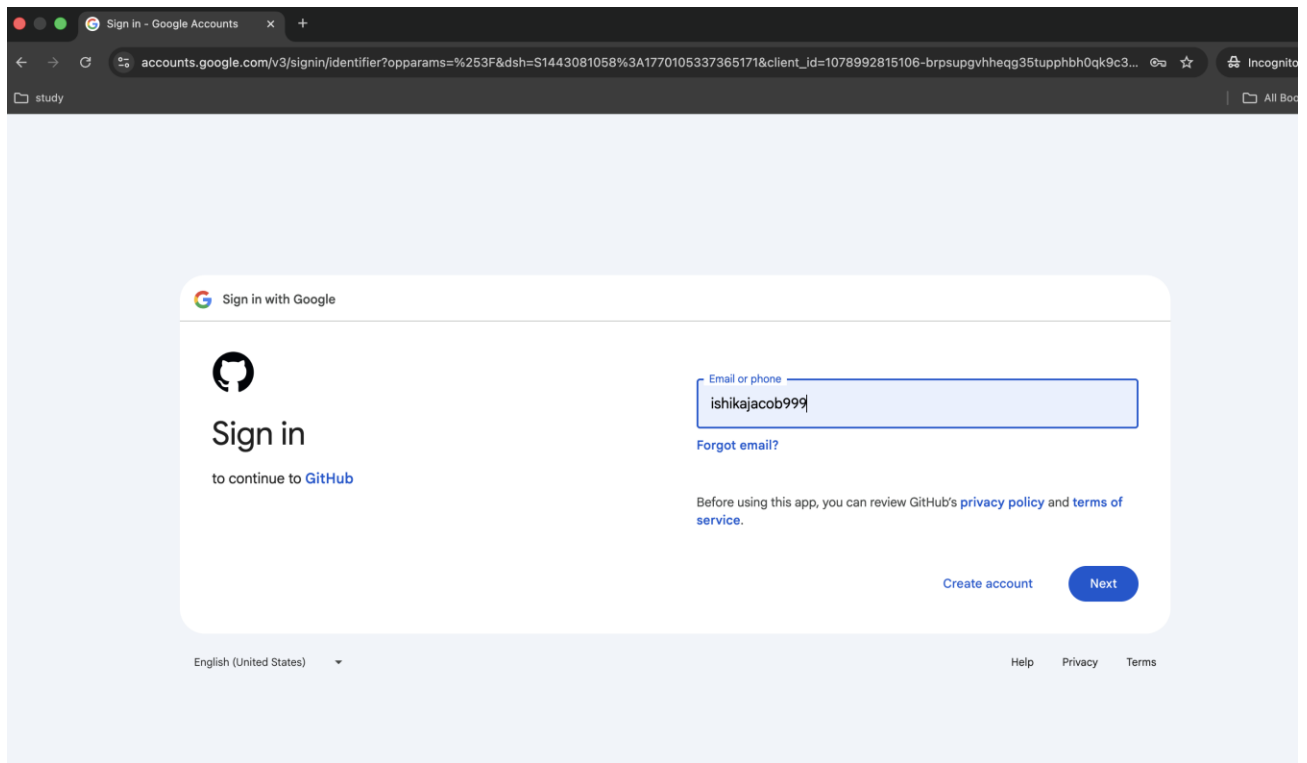
- **Select your country**
- **Click on create account**

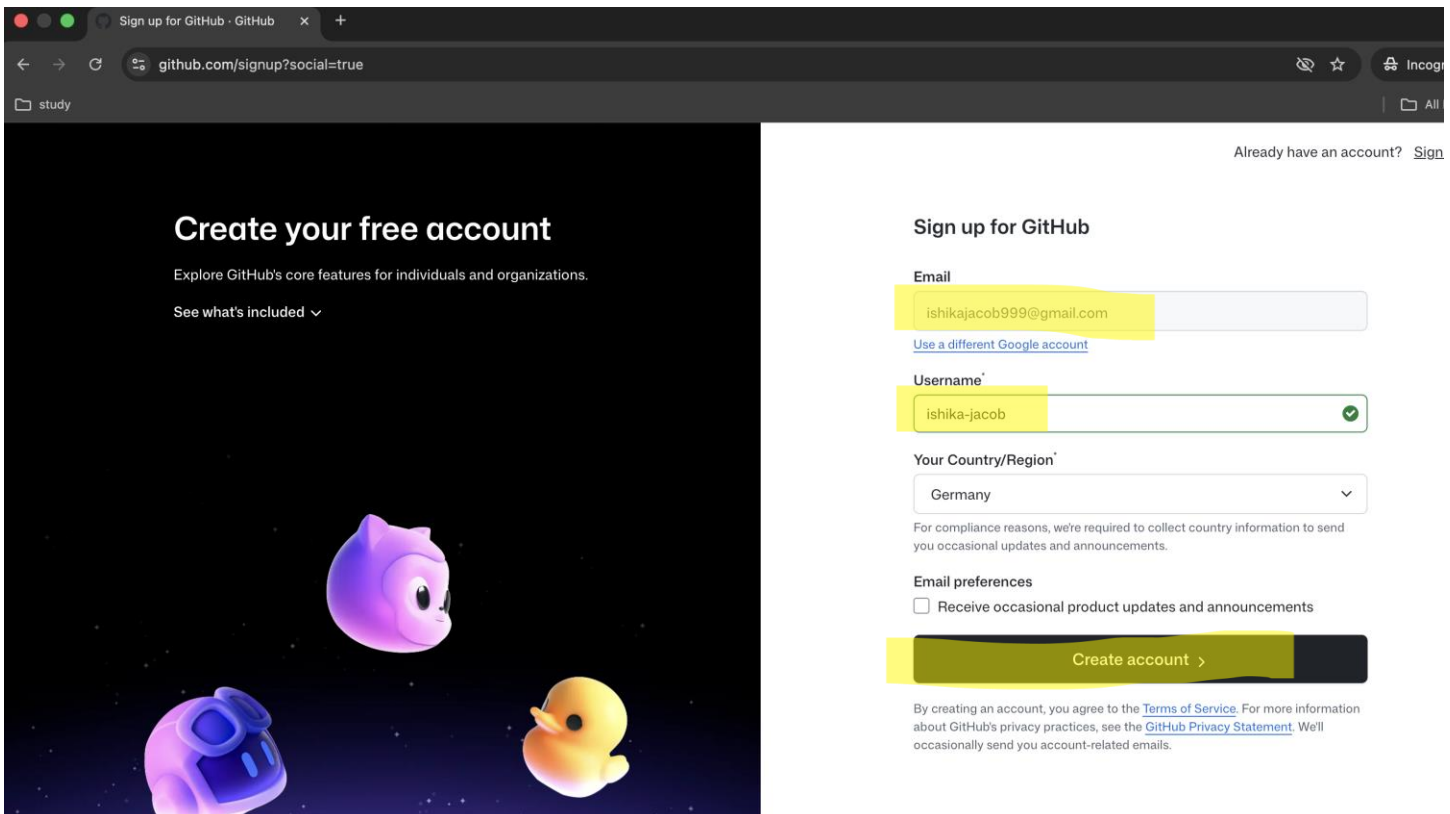
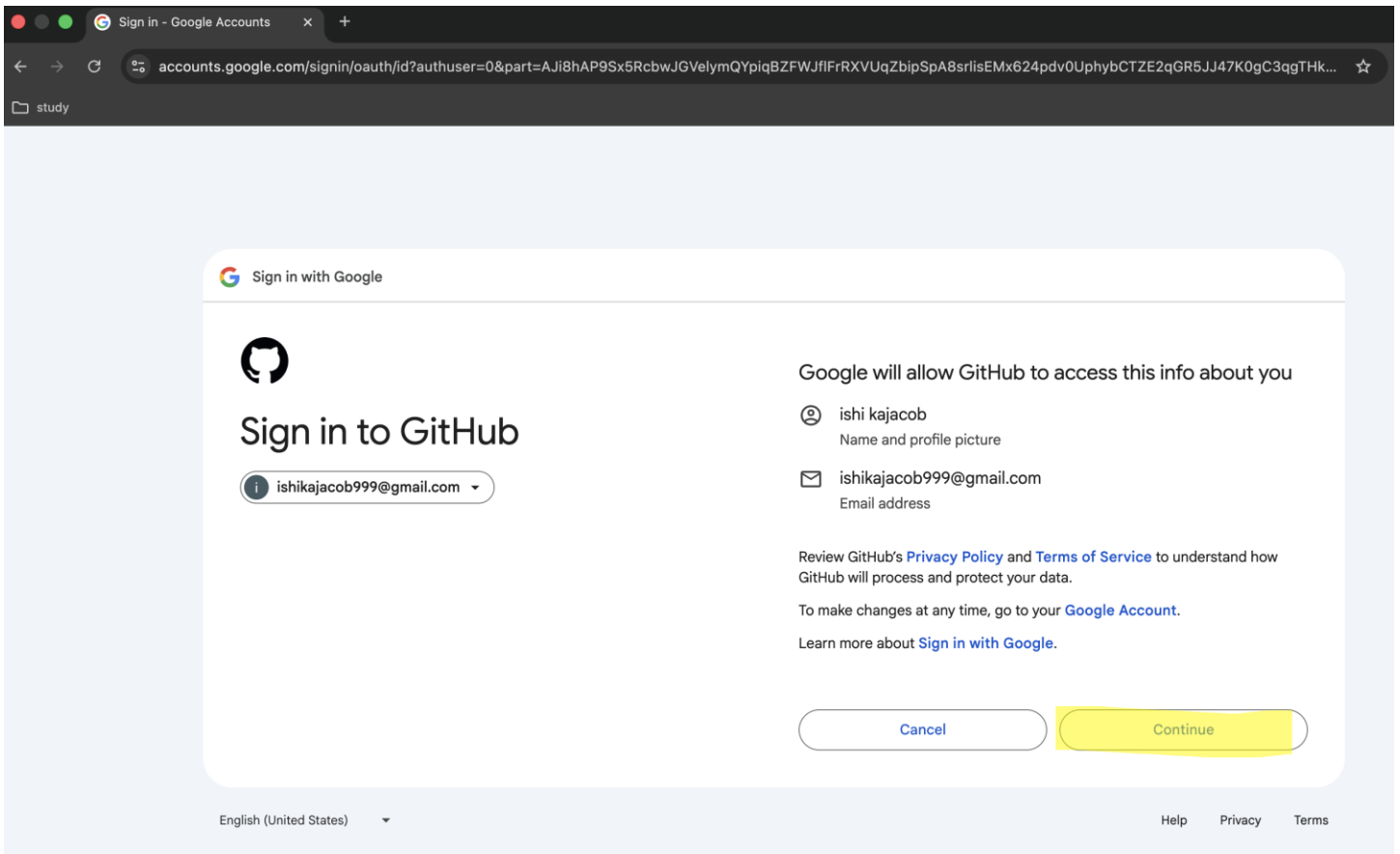
1. You can also use “Continue with Google option”



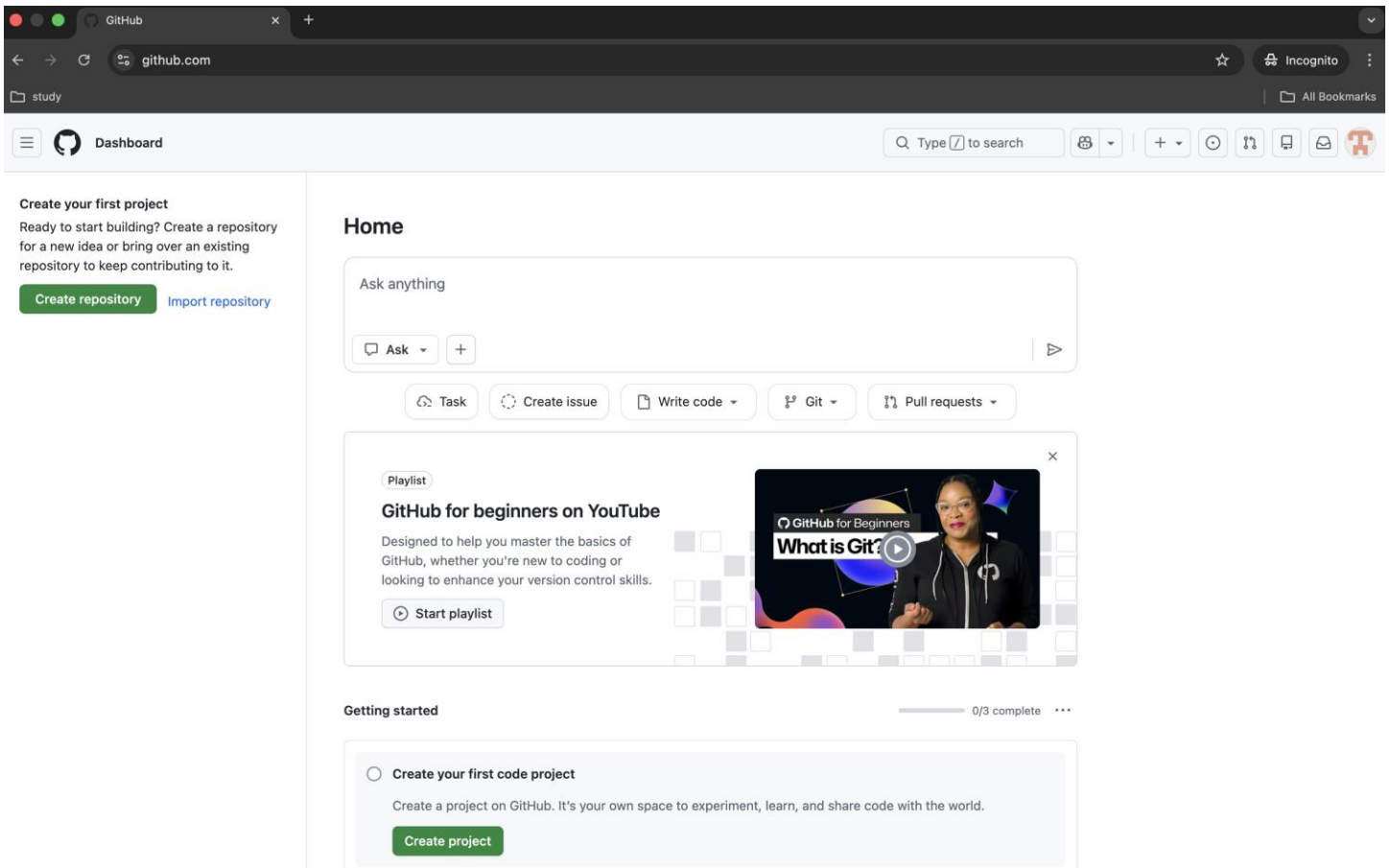
The screenshot shows the GitHub sign-up page in a web browser. The browser's address bar displays the URL: `github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home`. The page has a dark theme. On the left, a large section titled "Create your free account" includes the text "Explore GitHub's core features for individuals and organizations." and a link "See what's included". Below this is an illustration of three GitHub mascots (Octocat, Blob, and Duck) in space. On the right, the "Sign up for GitHub" form is visible. The "Continue with Google" button is highlighted with a yellow box. Below it is the "Continue with Apple" button. A separator "or" is present. The form includes input fields for "Email", "Password", and "Username", each with a small asterisk indicating it is required. Below the "Password" field, a note states: "Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter." Below the "Username" field, a note states: "Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen." The "Your Country/Region" field is a dropdown menu currently set to "Germany". Below this, a note states: "For compliance reasons, we're required to collect country information to send you occasional updates and announcements." The "Email preferences" section has a checkbox labeled "Receive occasional product updates and announcements", which is currently unchecked. At the bottom of the form is a dark button labeled "Create account >". Below the button, a small line of text states: "By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll

2. Give your email id and password:

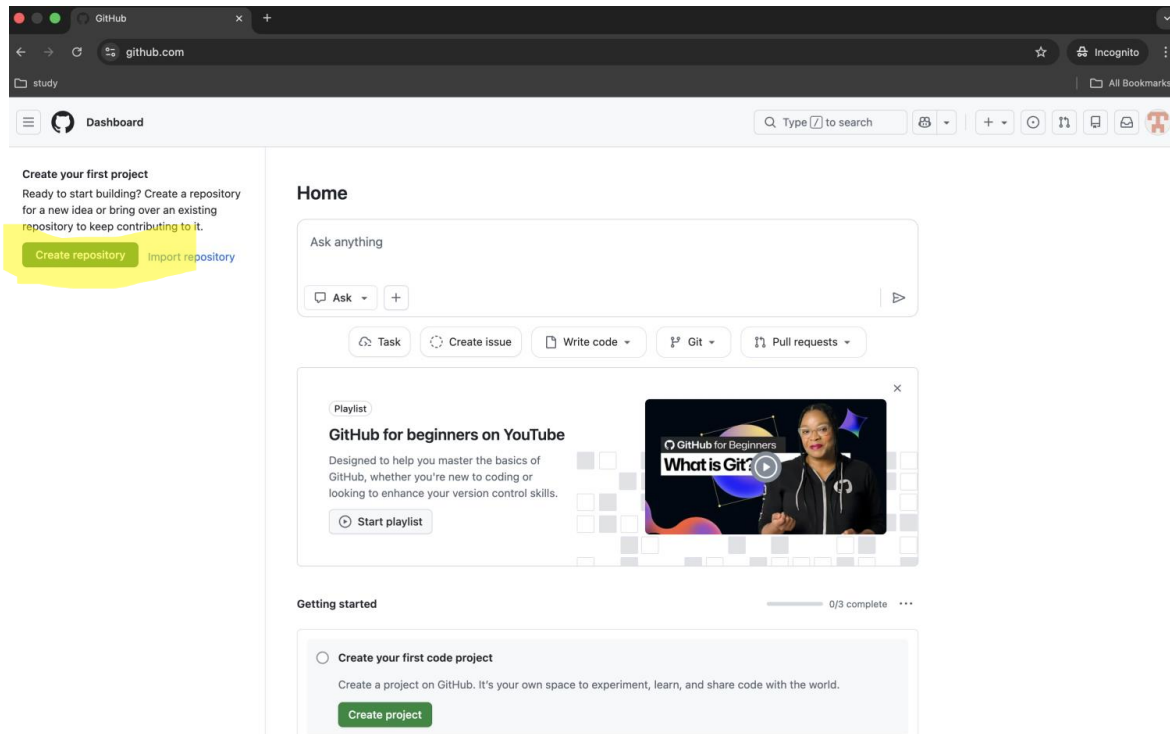




This is how the GitHub page look like once you create a GitHub account:



Now Click on “Create repository”



Give a repository name -> Select ->private and click on ->create repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner * ishika-jacob / Repository name *

Great repository names are short and memorable. How about [expert-spork?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility * Private

Choose who can see and commit to this repository

Add README Off

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license

Licenses explain how others can use your code. [About licenses](#)

[Create repository](#)

This is how the Github page look like when you create a GitHub repository :

AI-Integration_day_1_lab_1 Private

Watch 0 Fork 0 Star 0

Set up GitHub Copilot

Use GitHub's AI pair programmer to autocomplete suggestions as you code.

[Get started with GitHub Copilot](#)

Add collaborators to this repository

Search for people using their GitHub username or email address.

[Invite collaborators](#)

Quick setup — if you've done this kind of thing before

[Set up in Desktop](#) or [HTTPS](#) [SSH](#)

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# AI-Integration_day_1_lab_1" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git
git branch -M main
git push -u origin main
```

3. Optional (but recommended)

- Add profile picture
- Add bio:
"AI Consultant in training"

✅ Done

STEP 3 — Install GitHub Extension in VSCode

1. Open Extensions

- In VSCode press:
 - **Windows/Linux:** Ctrl + Shift + X
 - **Mac:** Cmd + Shift + X

2. Install

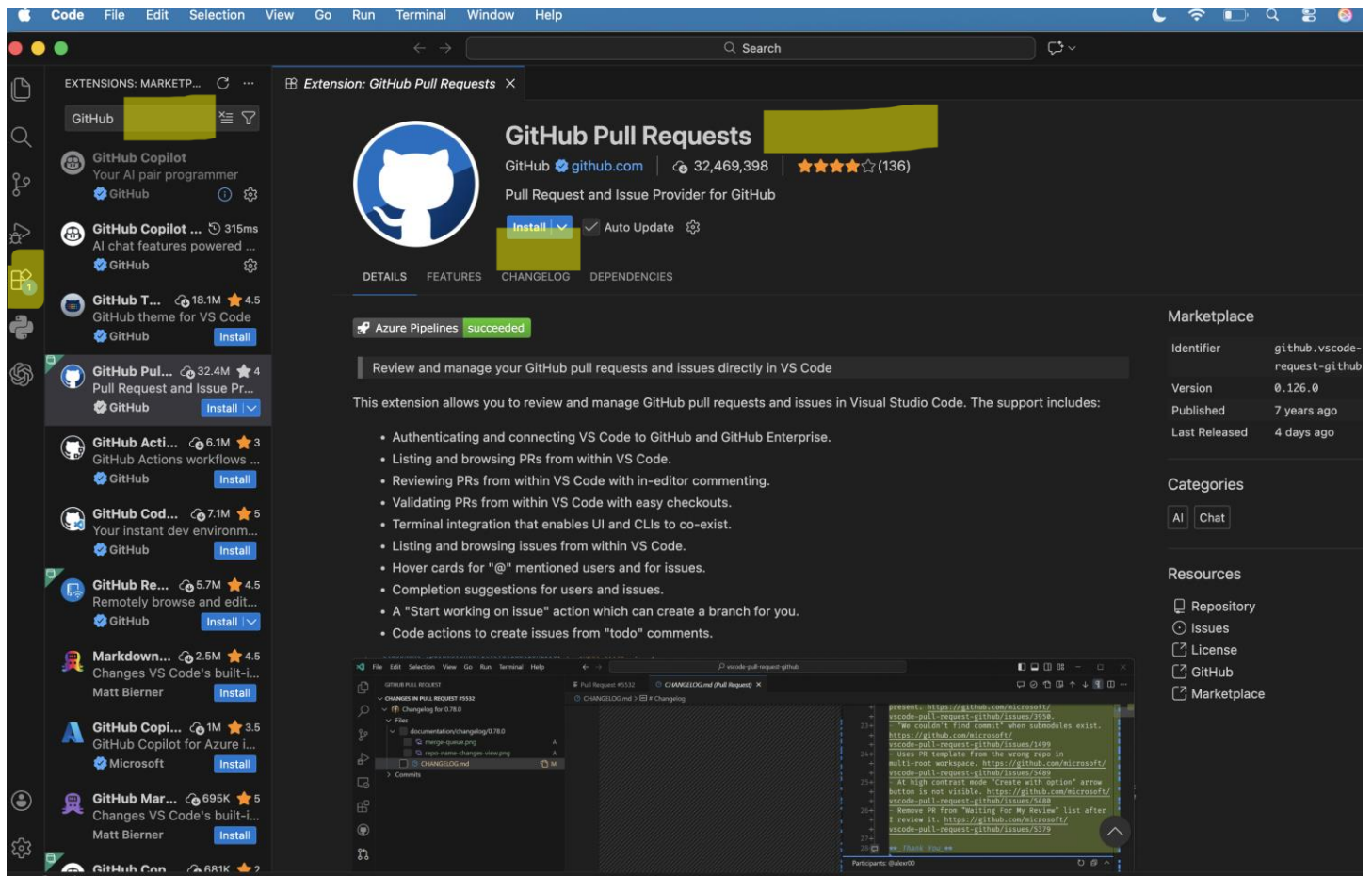
- Search: **GitHub**
- Install the **official GitHub extension** (by GitHub)

3. Sign in

- Click **Accounts icon** (bottom-left)
- Choose **Sign in with GitHub**
- Browser opens → authorize VSCode

✅ VSCode ↔ GitHub connected

Steps with screenshot: click on Extension in the left corner -> Type GitHub in the search box -> select GitHub Pull Requests -> Click on install



✓ Select and install:

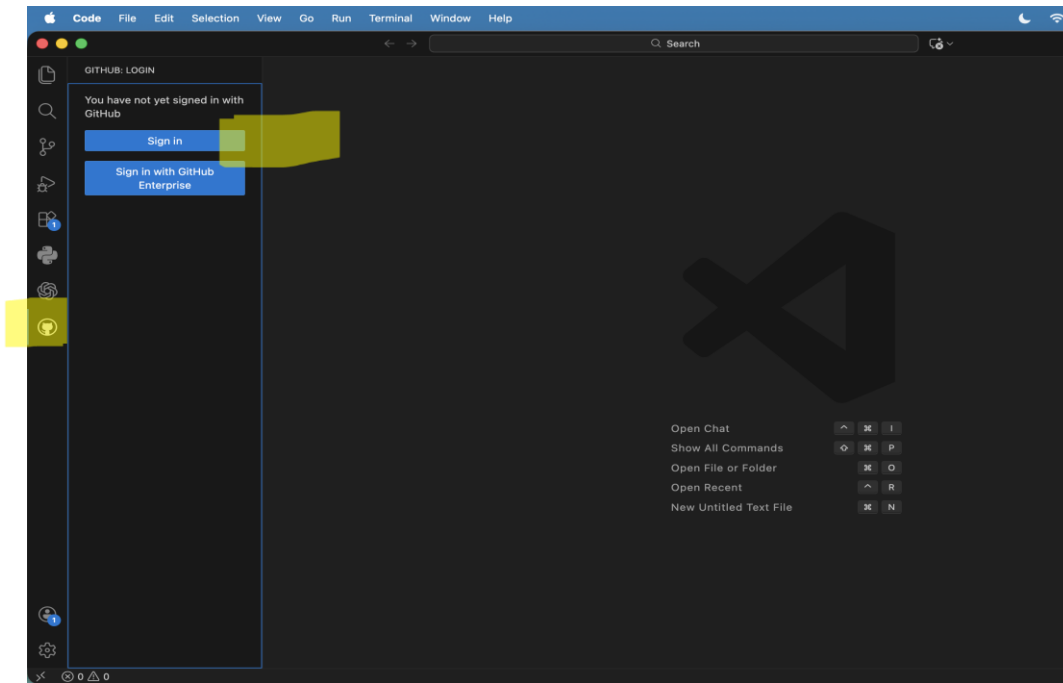
GitHub Pull Requests

- Publisher: **GitHub**
- Description: *Pull Request and Issue Provider for GitHub*
- Millions of installs ✓

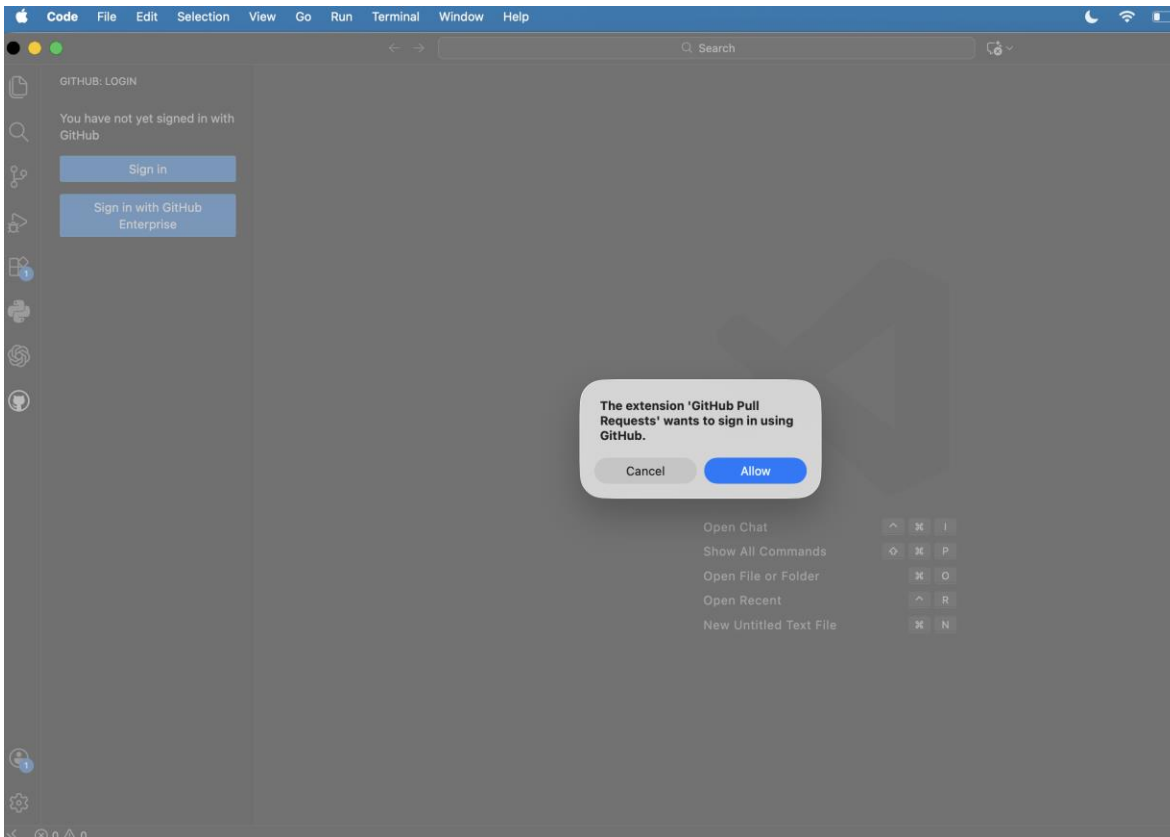
Go ahead and click **Install**.

This extension:

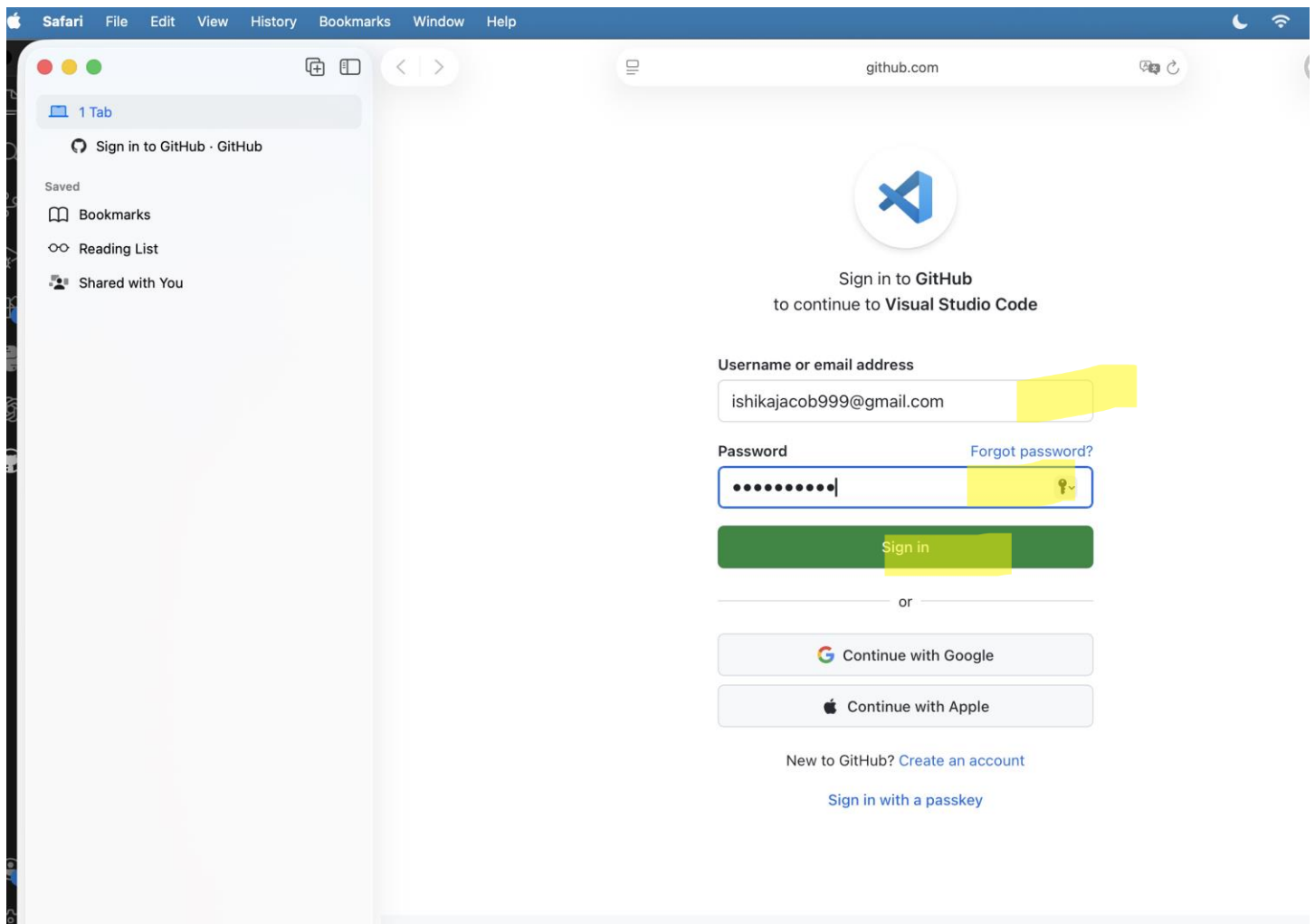
- Let's VSCode **sign in to GitHub**
- Enables pushing / pulling repositories
- Integrates GitHub directly into VSCode



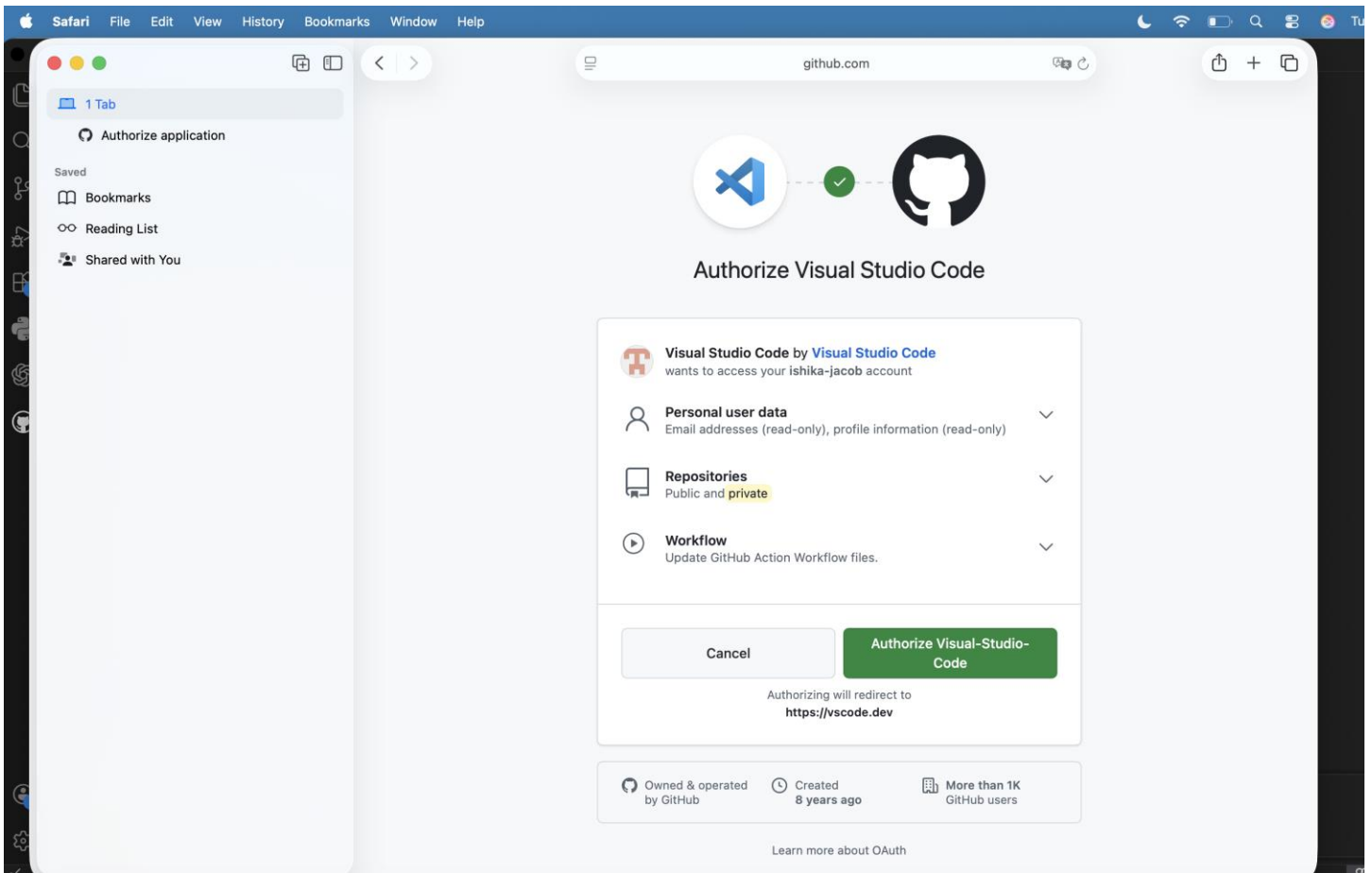
Click on “Allow”



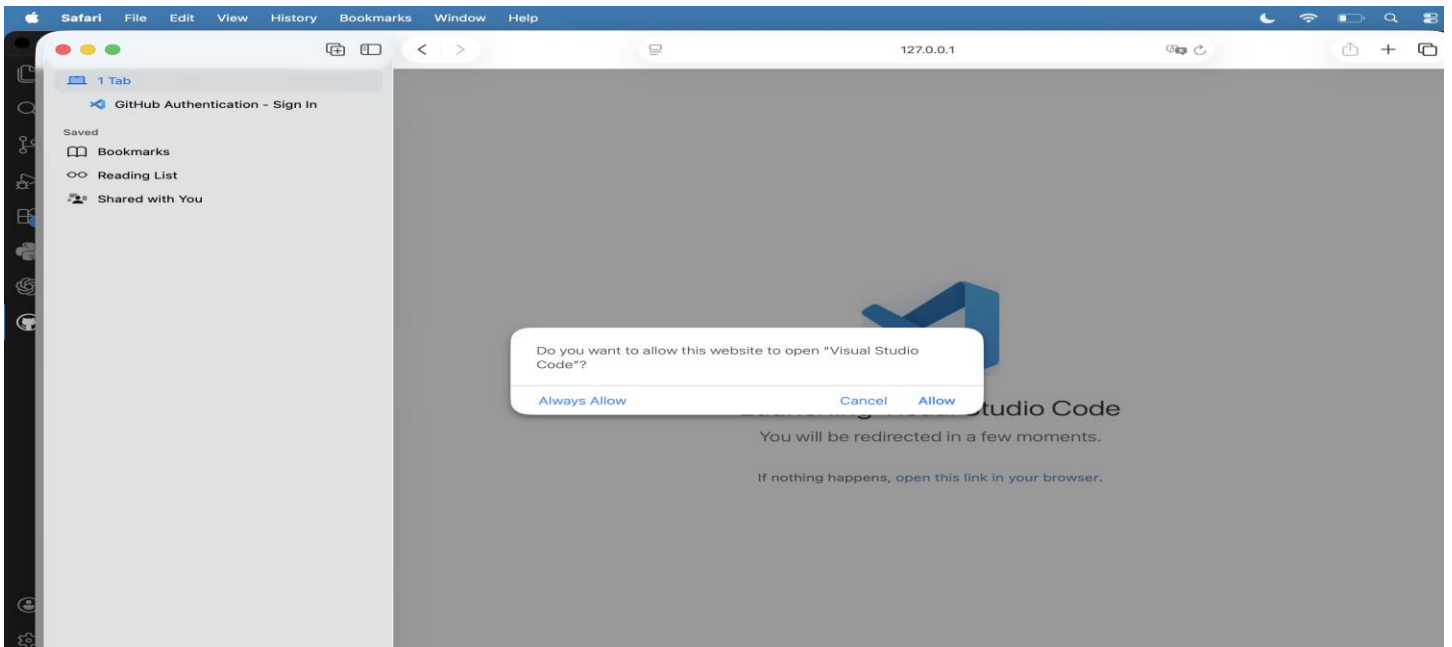
Enter your GitHub email id and password and click on “sign in”:



Click on “Authorize VSCode”



Click on "Allow"



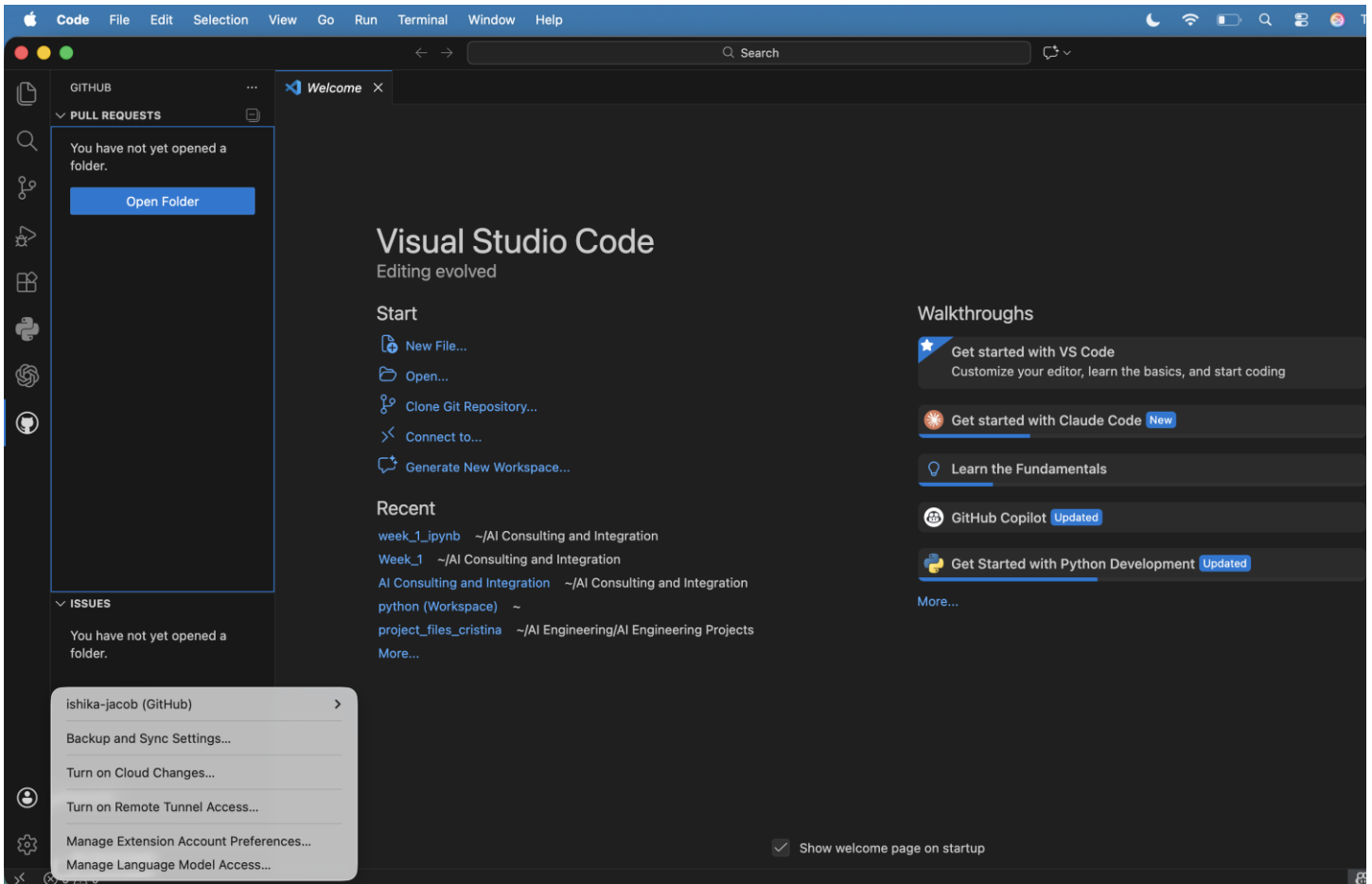
Click on GitHub to check your GitHub Account is visible:

 **What happens next**

After installing:

1. Look at the **bottom-left corner** of VSCode
2. Click the **Accounts icon**
3. Choose “**Sign in with GitHub**”
4. Complete the browser login

Once that’s done, you’re ready for **Configure Git**.

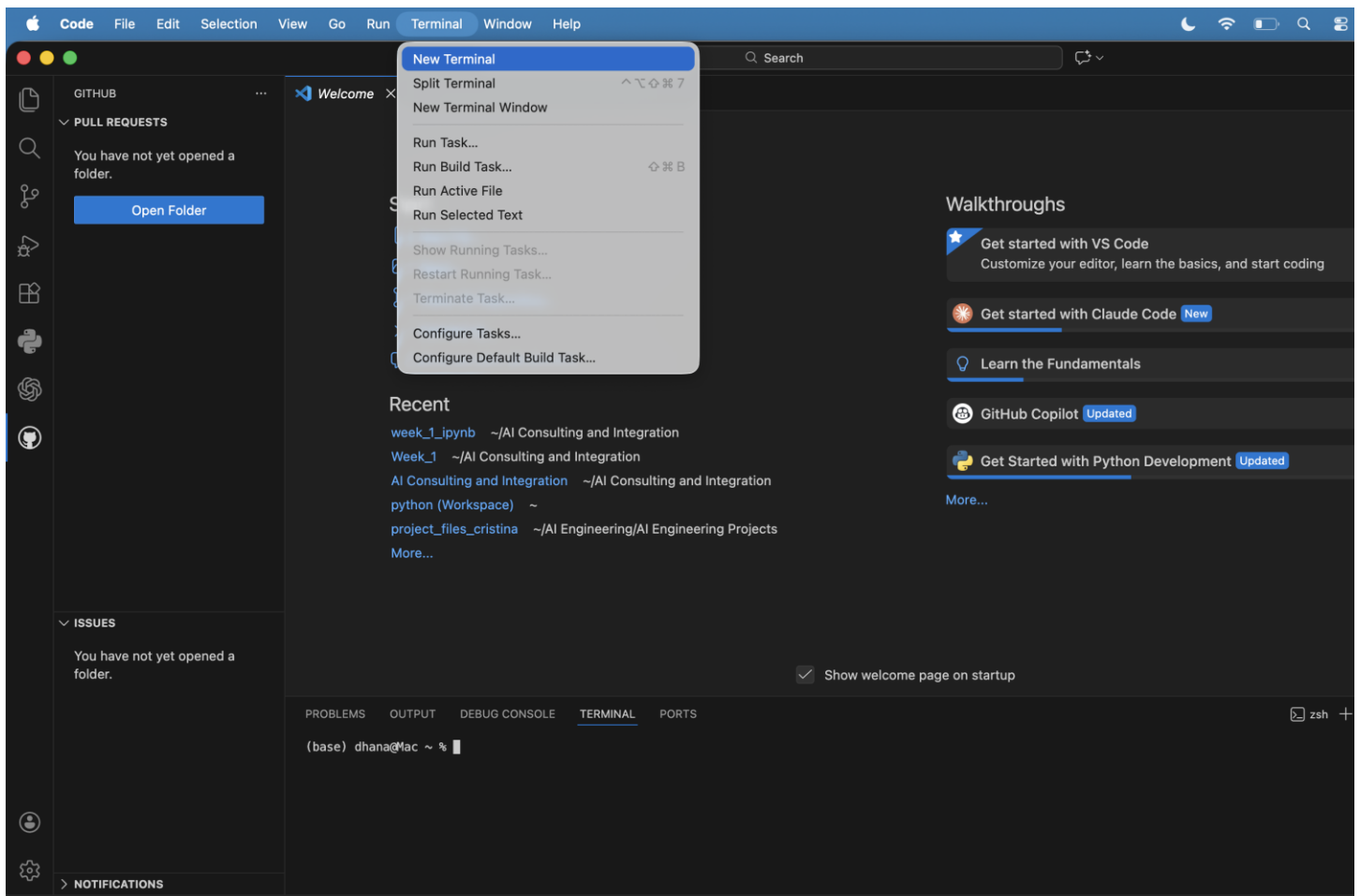


STEP 4 — Configure Git (⚠ VERY IMPORTANT)

This step is **non-negotiable**. Without it, commits WILL FAIL.

1. Open VSCode Terminal

- Press: ``Ctrl + ``` (backtick)
- Or: **Terminal** → **New Terminal**

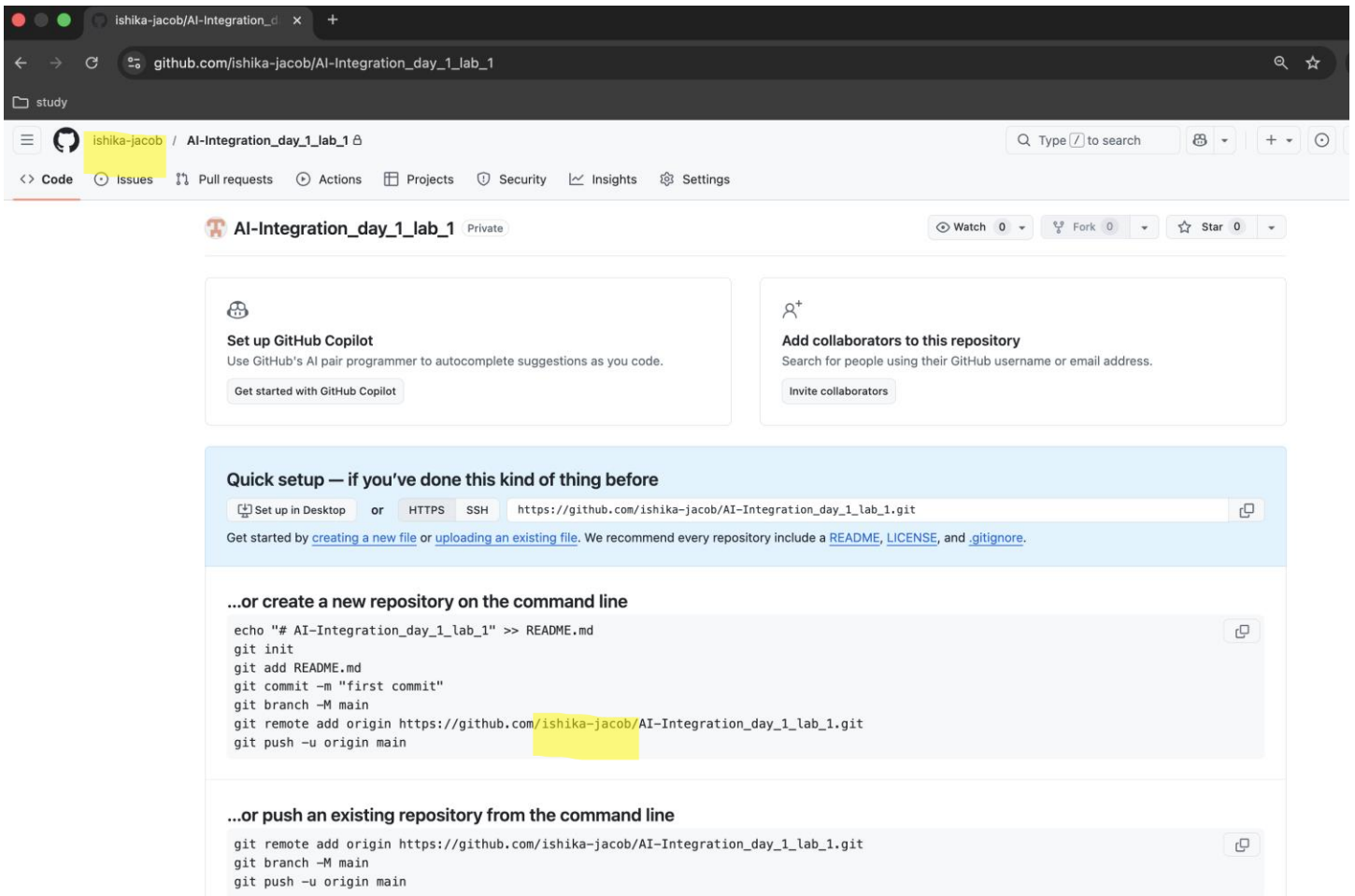


2. Set your name

`git config --global user.name "Your Full Name"`

Example: `git config --global user.name "Jane Doe"`

Example: `git config --global user.name "ishika-jacob"`



3. Set your email (MUST match GitHub)

`git config --global user.email "your.email@example.com"`

Example : `git config --global user.email "ishikajacob999@example.com"`

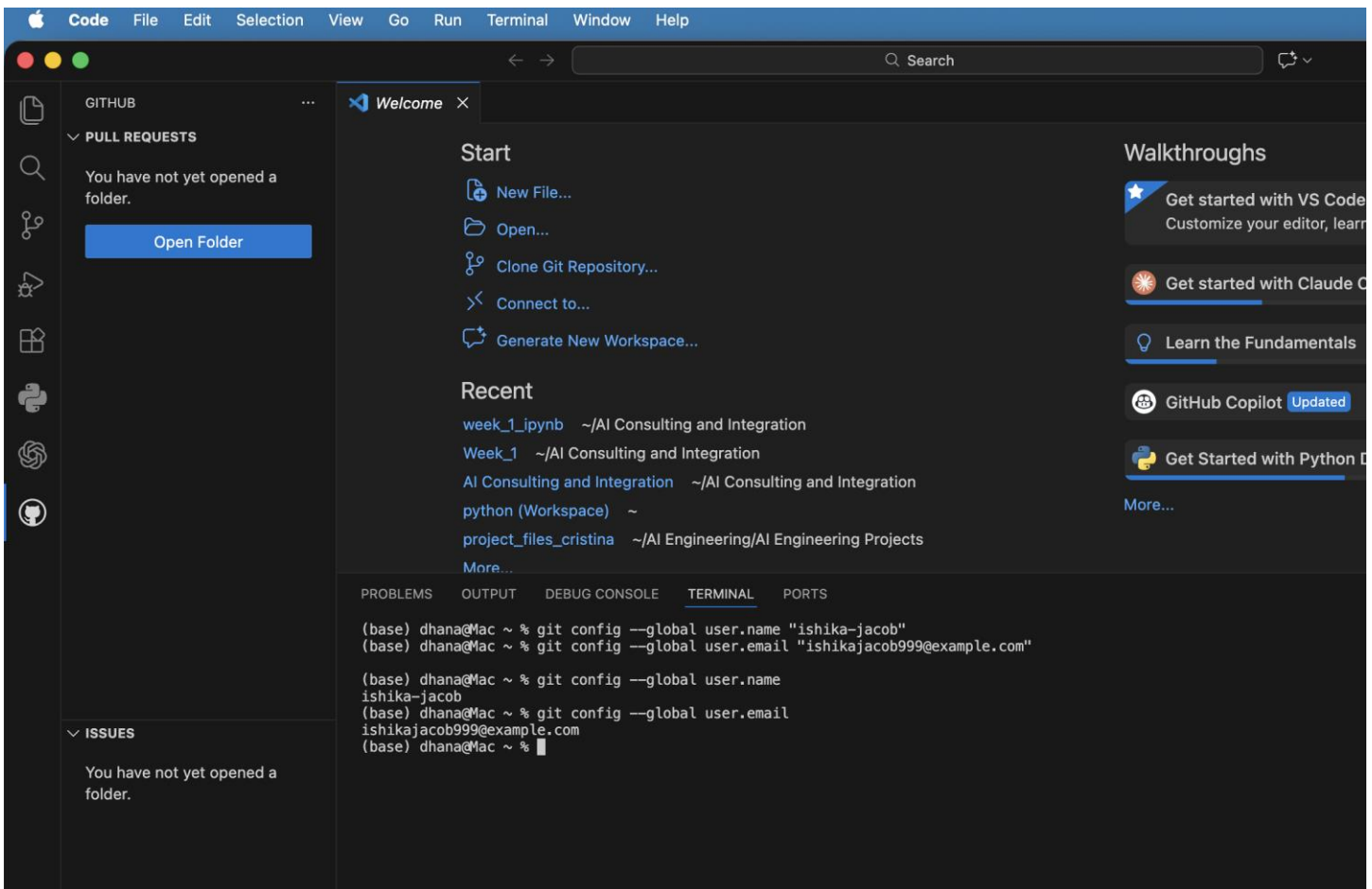
4. Verify

`git config --global user.name`

`git config --global user.email`

You should see your name + email printed.

✅ Git is now correctly configured



STEP 6 — Install Conda (Python Environment Manager)

1. Download (recommended)

- 👉 <https://docs.conda.io/en/latest/miniconda.html>

2. Install

- Run installer
- ⚠️ **CHECK:** “Add Conda to PATH”
- Finish install

3. Verify in VSCode

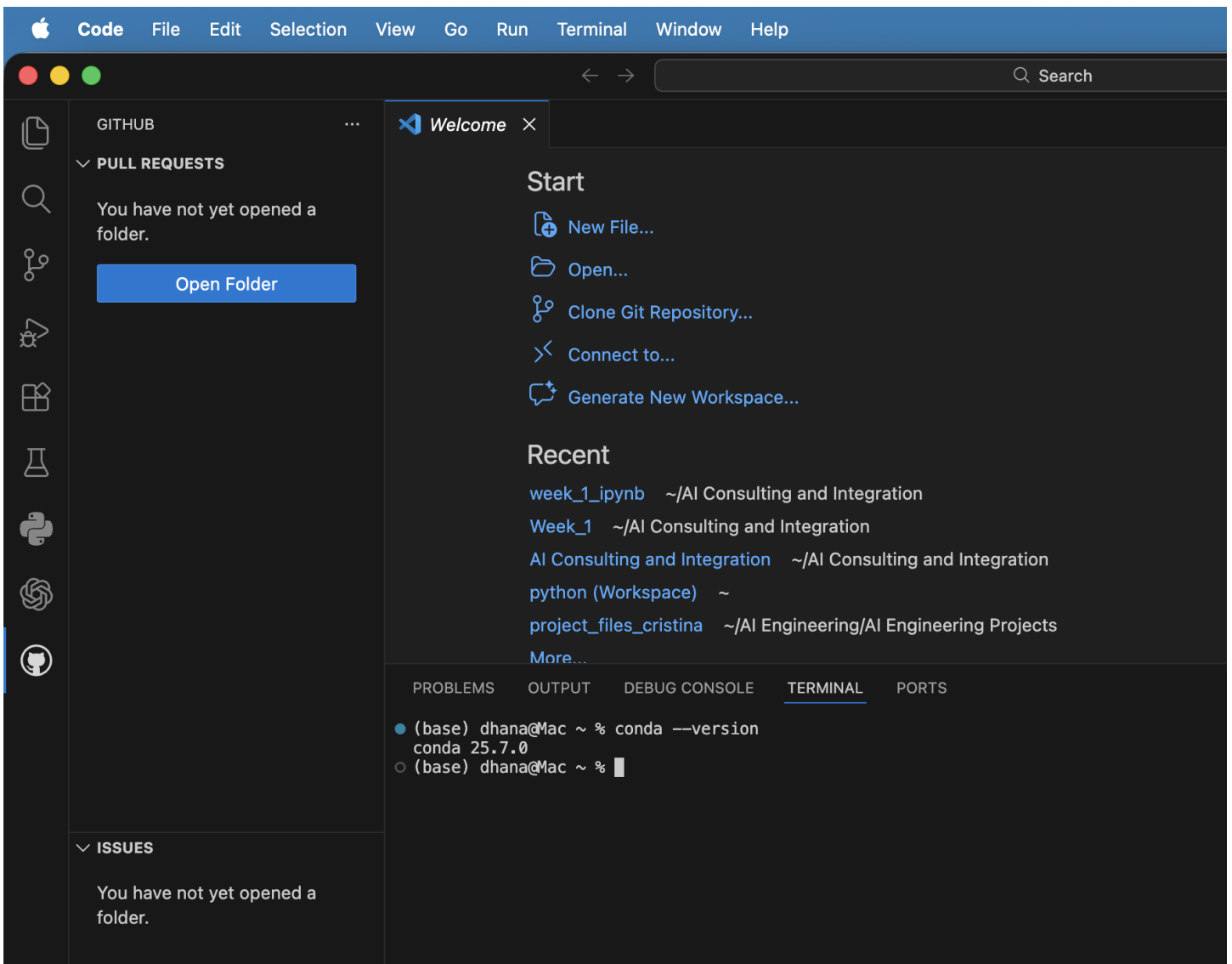
Open a **new terminal** and run:

```
conda --version
```

You should see something like below or the other conda versions as example:

```
conda 23.x.x
```

Example :



Conda installed correctly

STEP 7 — Create a Conda Environment (VSCode UI)

1. Open Command Palette

- Ctrl + Shift + P / Cmd + Shift + P

2. Select Interpreter

- Type: **Python: Select Interpreter**
- Click **Create Environment**

3. Choose

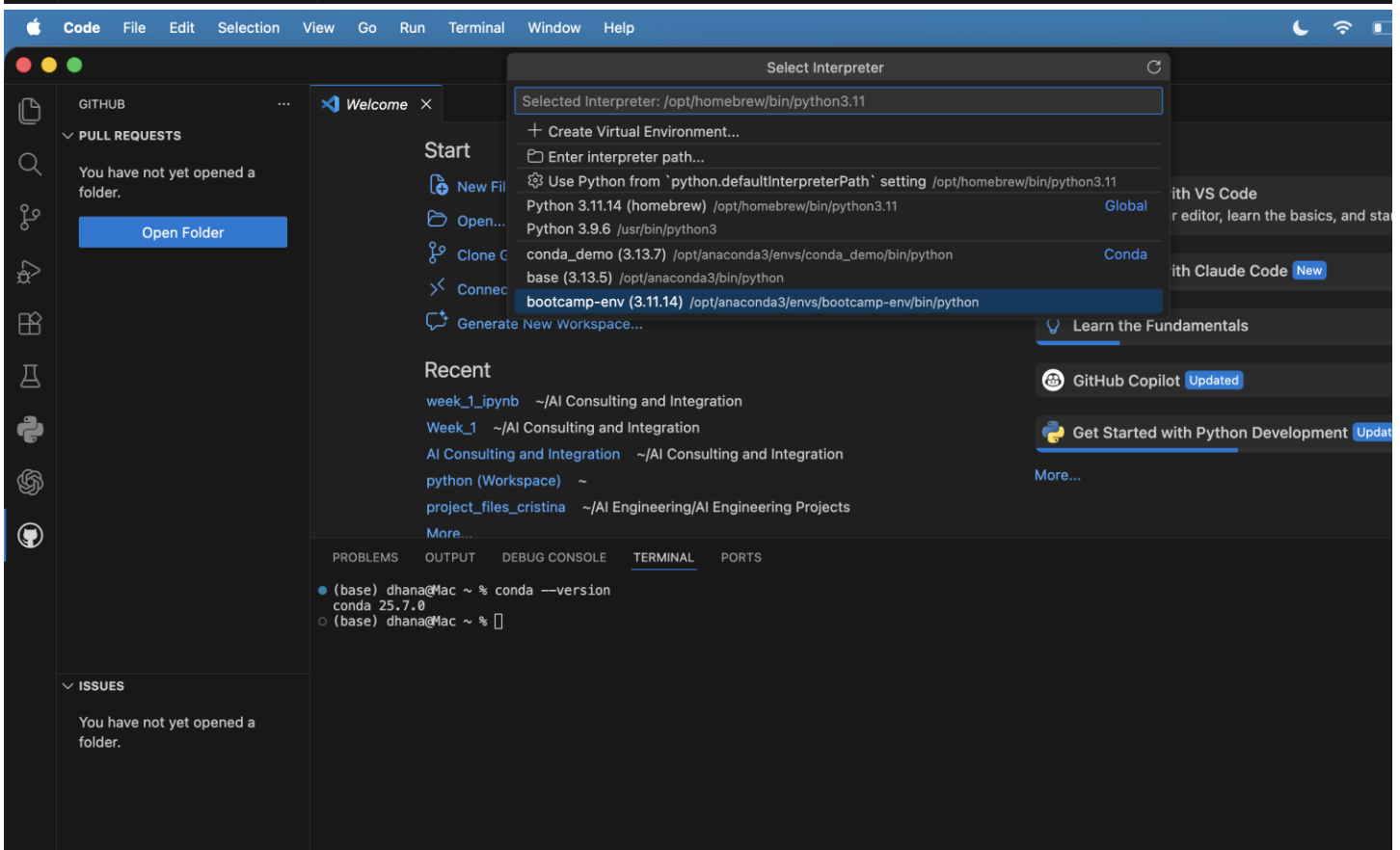
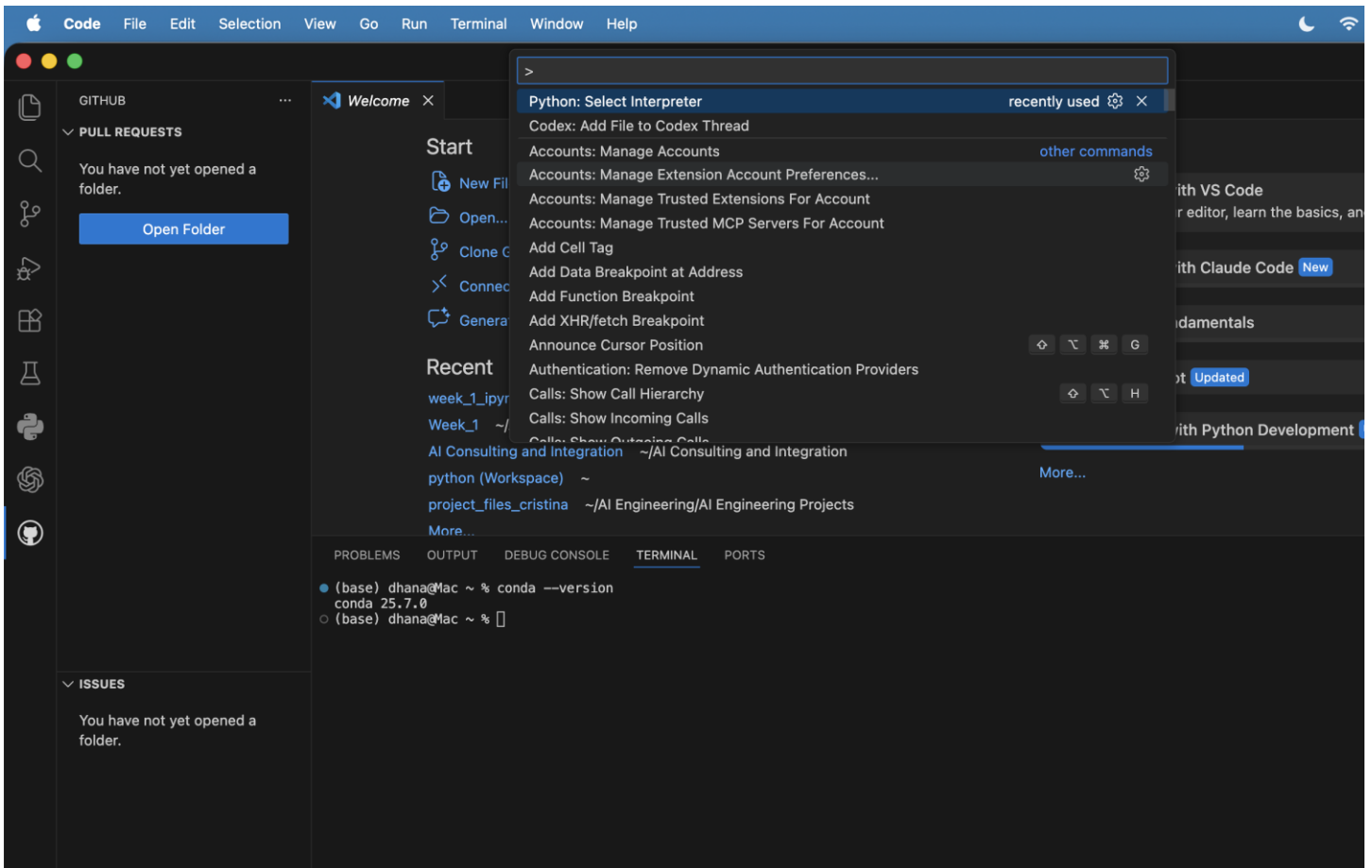
- Environment type: **Conda**
- Python version: **3.11**

- Name: bootcamp-env

 Wait 1–2 minutes

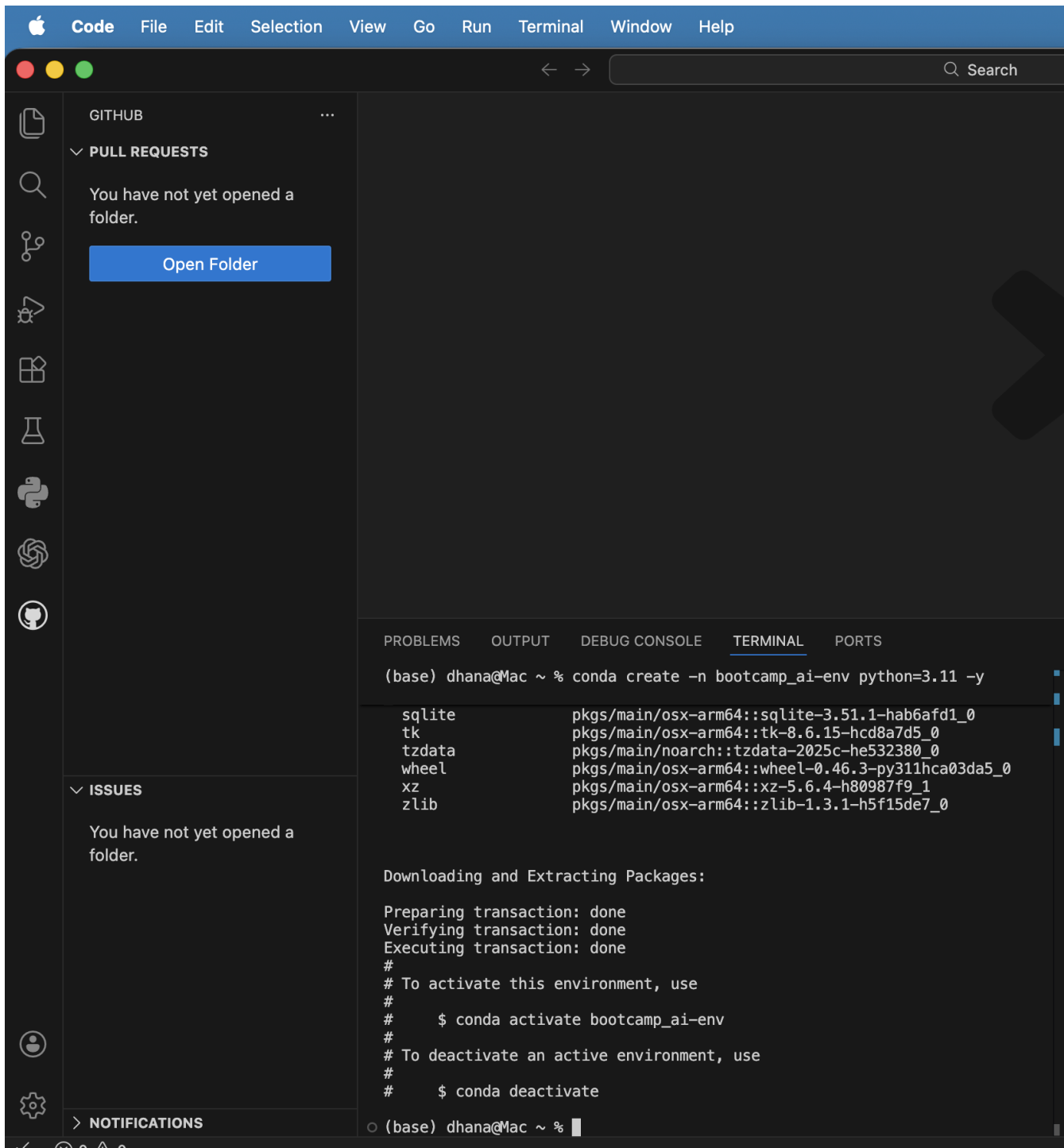
4. Verify

- Bottom-right corner of VSCode should show:

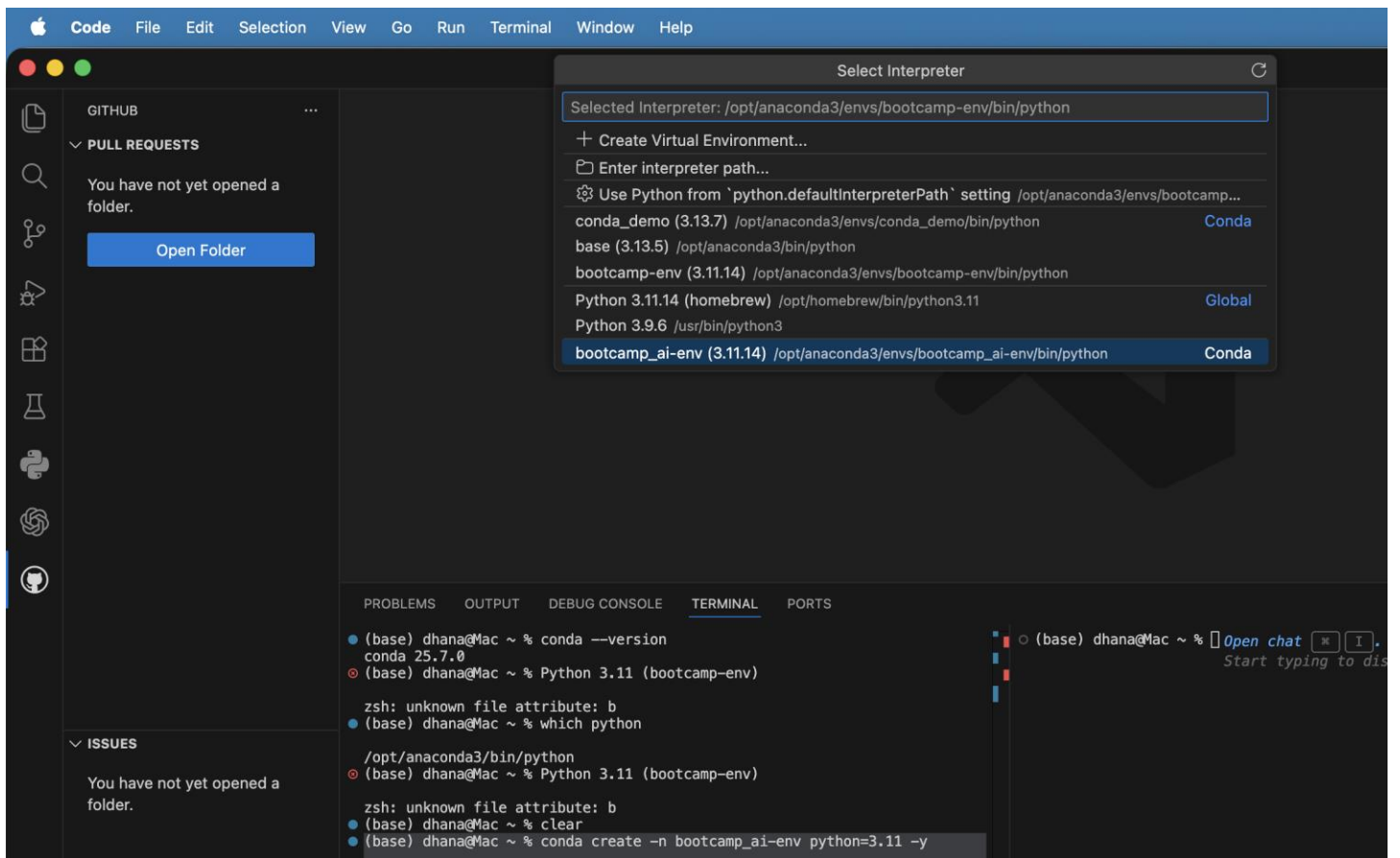


Create a new environment -> Click on terminal-> New Terminal:

conda create -n bootcamp_ai-env python=3.11 -y



Click on cmd+shift+P on macos or ctrl+shift+P ->Select Interpreter ->select bootcamp_ai-env



STEP 8 — Install Jupyter Extension

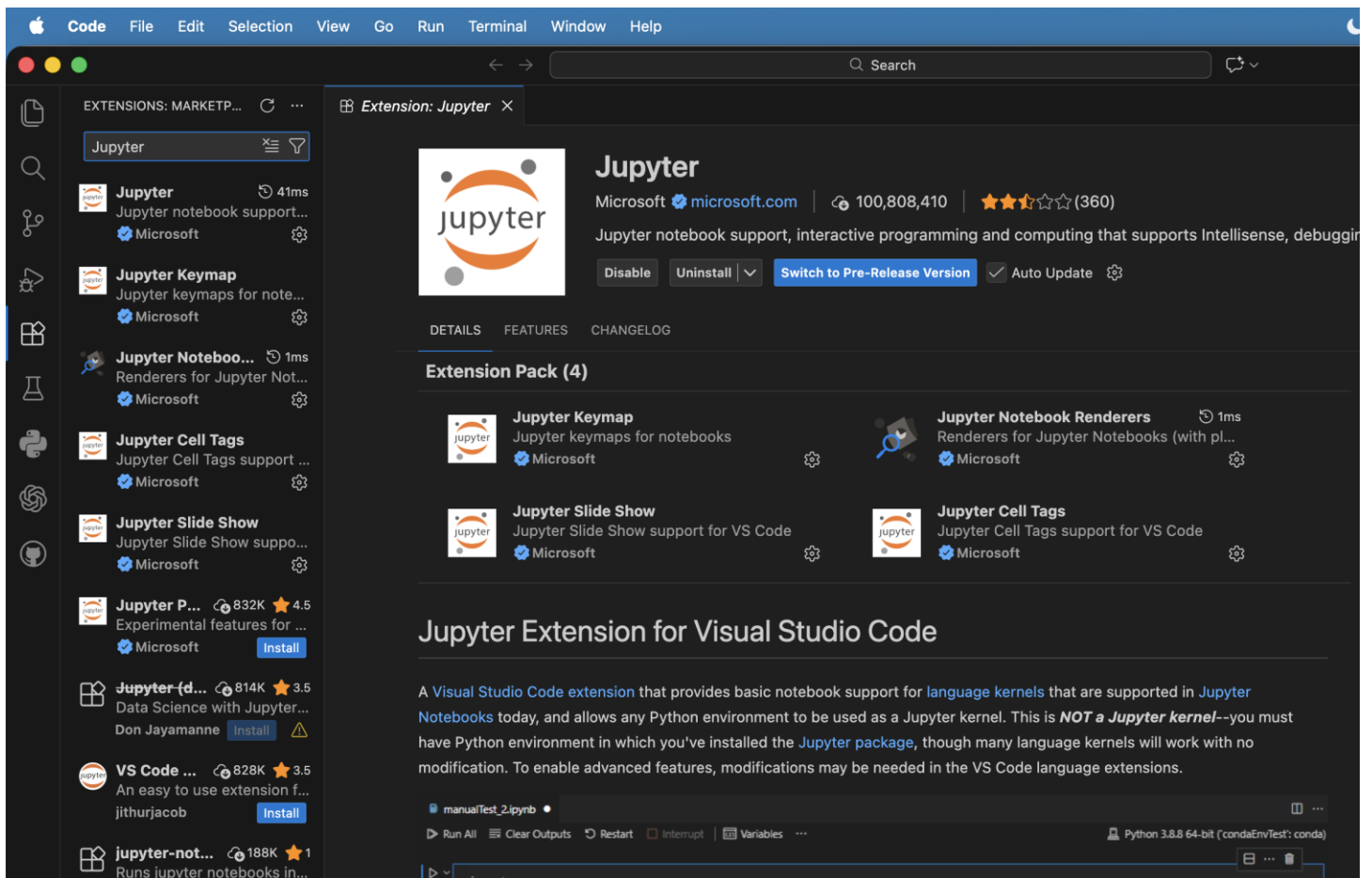
1. Open Extensions

- Ctrl + Shift + X

2. Search

- **Jupyter**
- **Install Jupyter (Microsoft)**

✅ Installed



STEP 9 — Create & Run a Jupyter Notebook

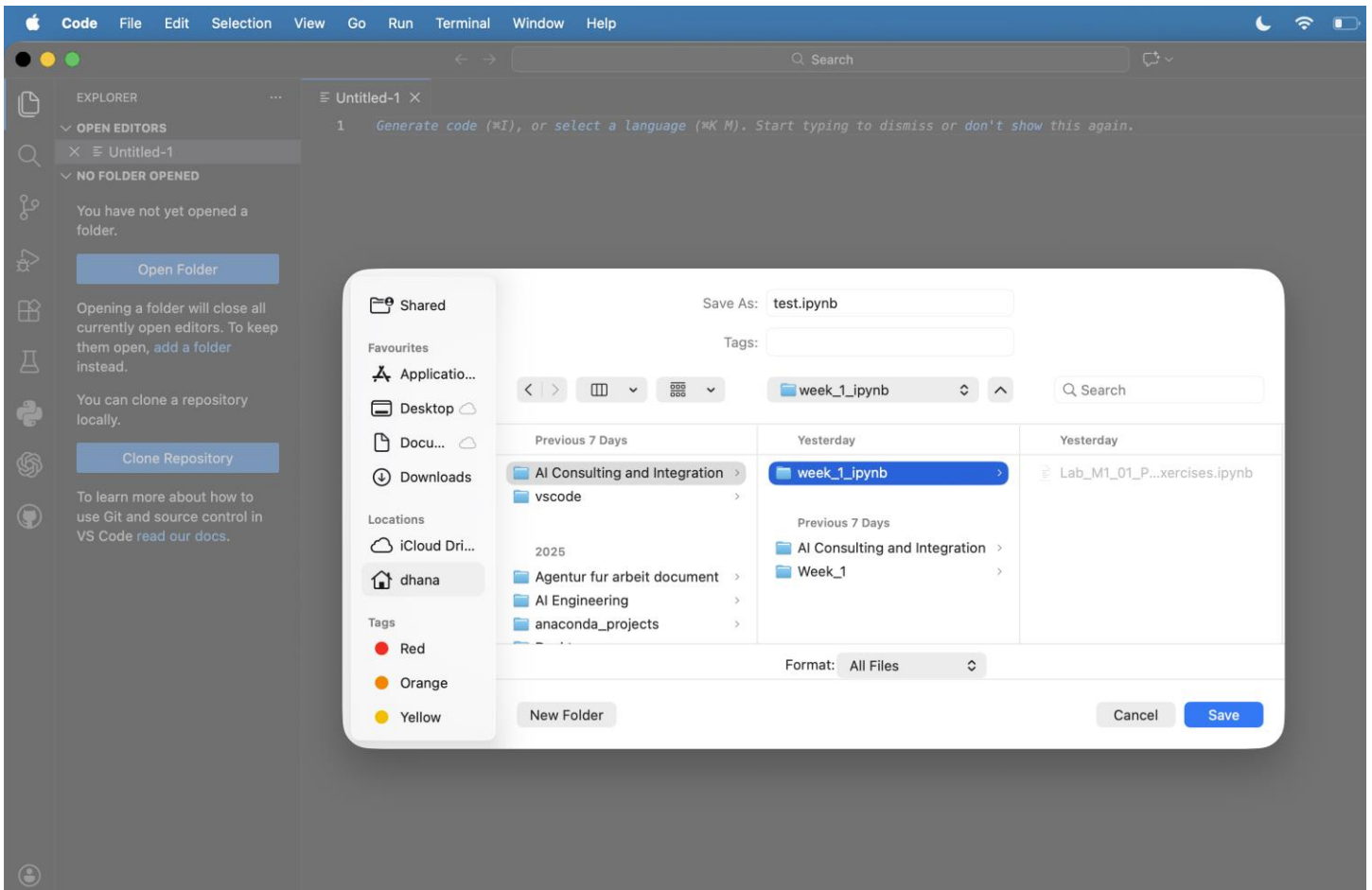
1. Create Notebook

- File → New File
- Save as: test.ipynb

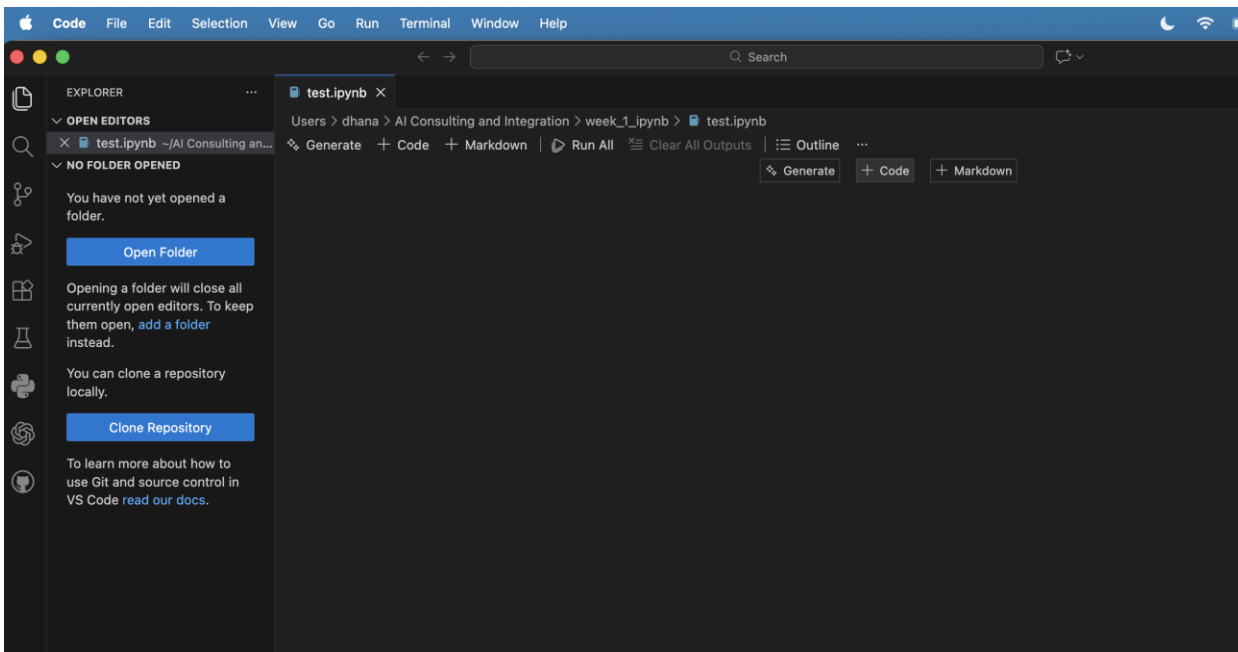
2. Select Kernel (CRITICAL)

- Top-right → Select Kernel
- Choose: **bootcamp_ai-env**

3. Test Code



Click on -> code



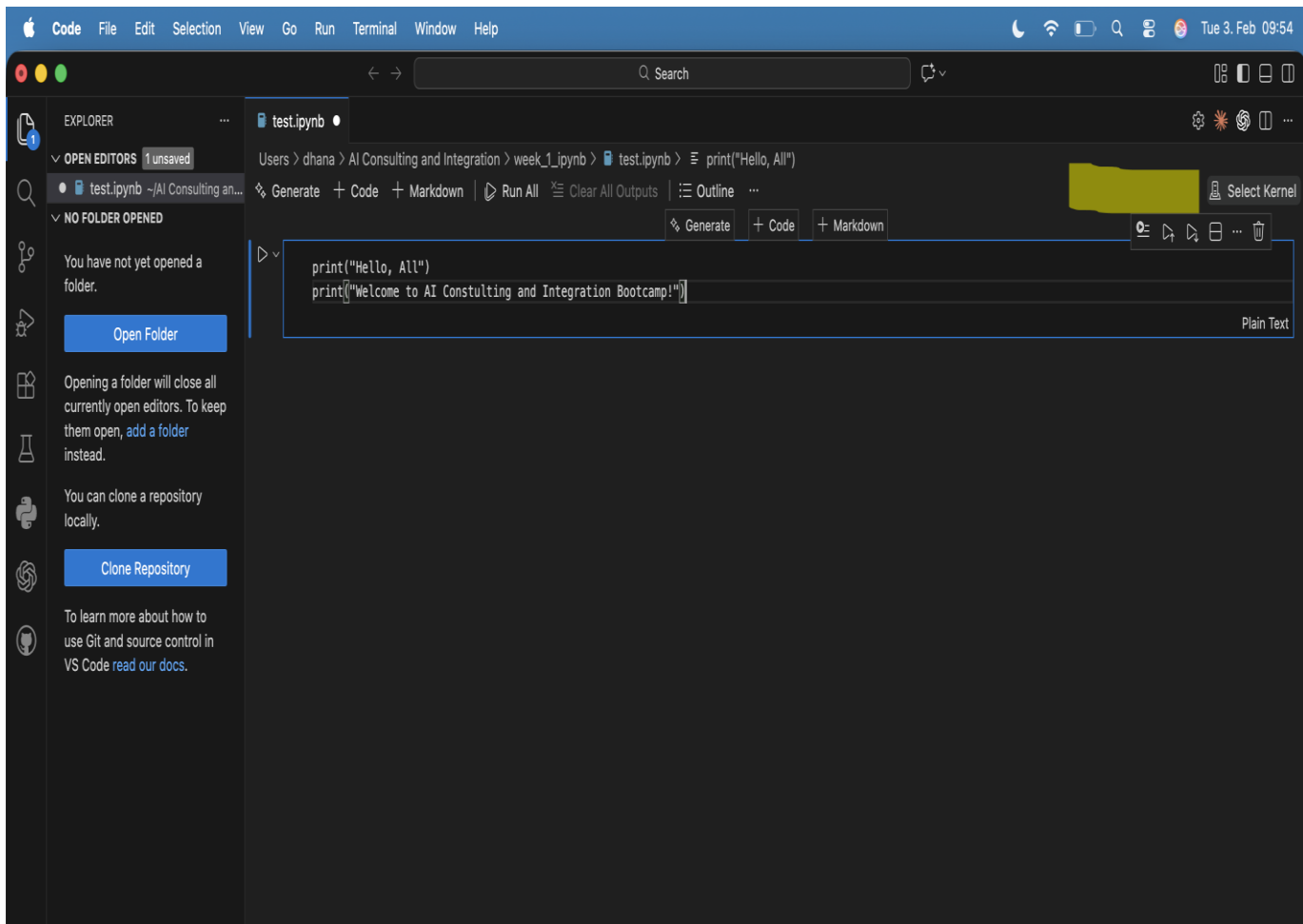
Give sample

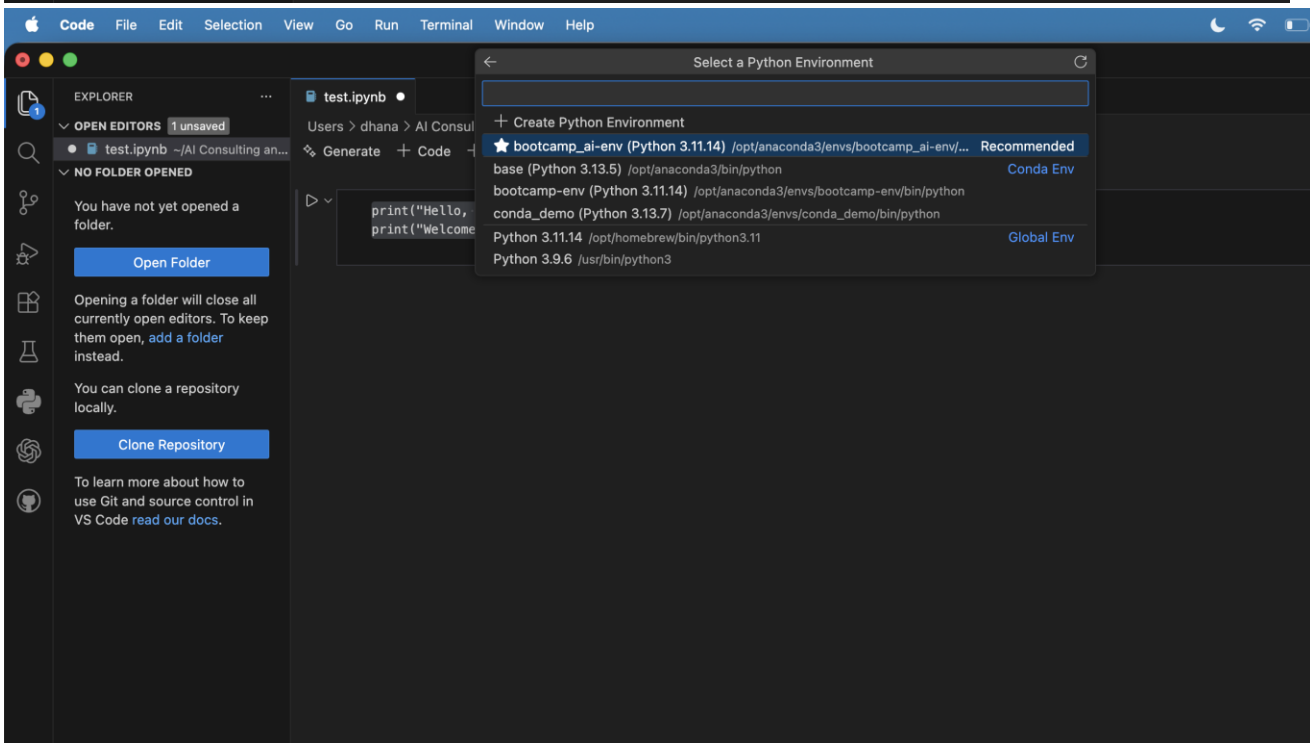
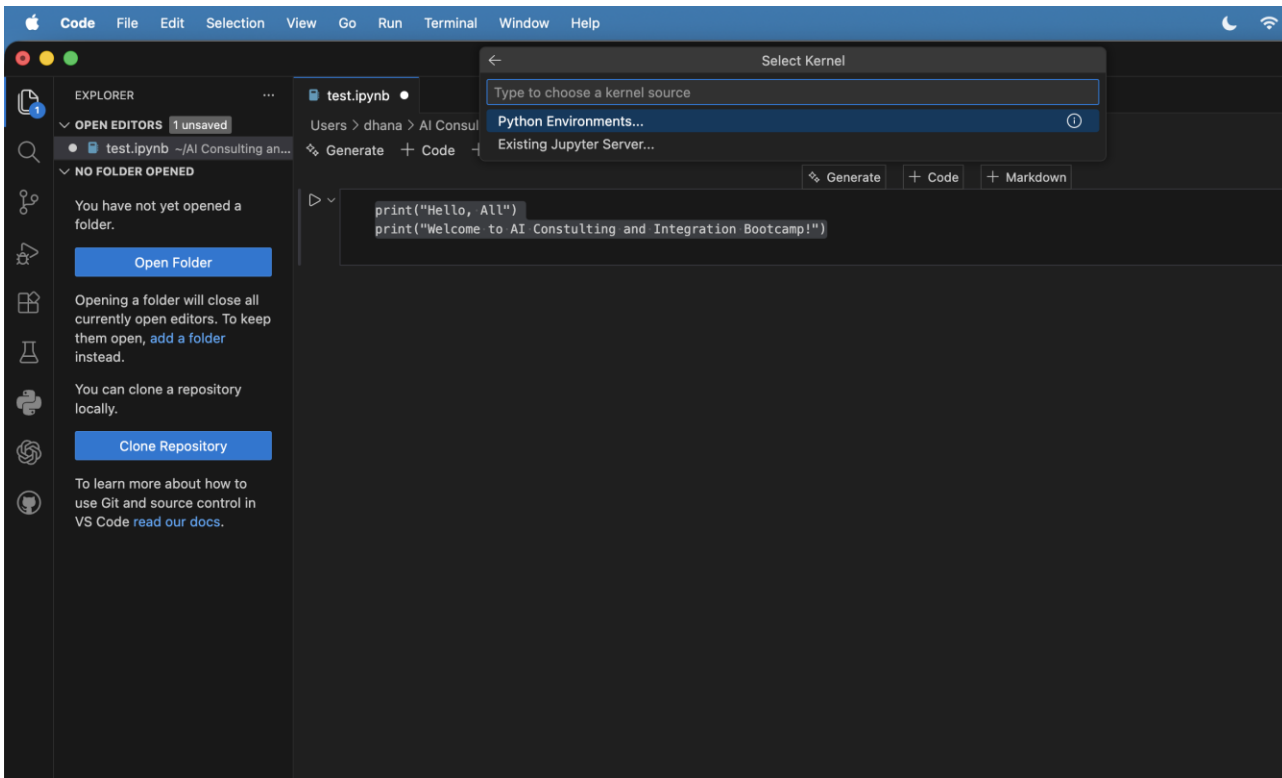
test code in code box

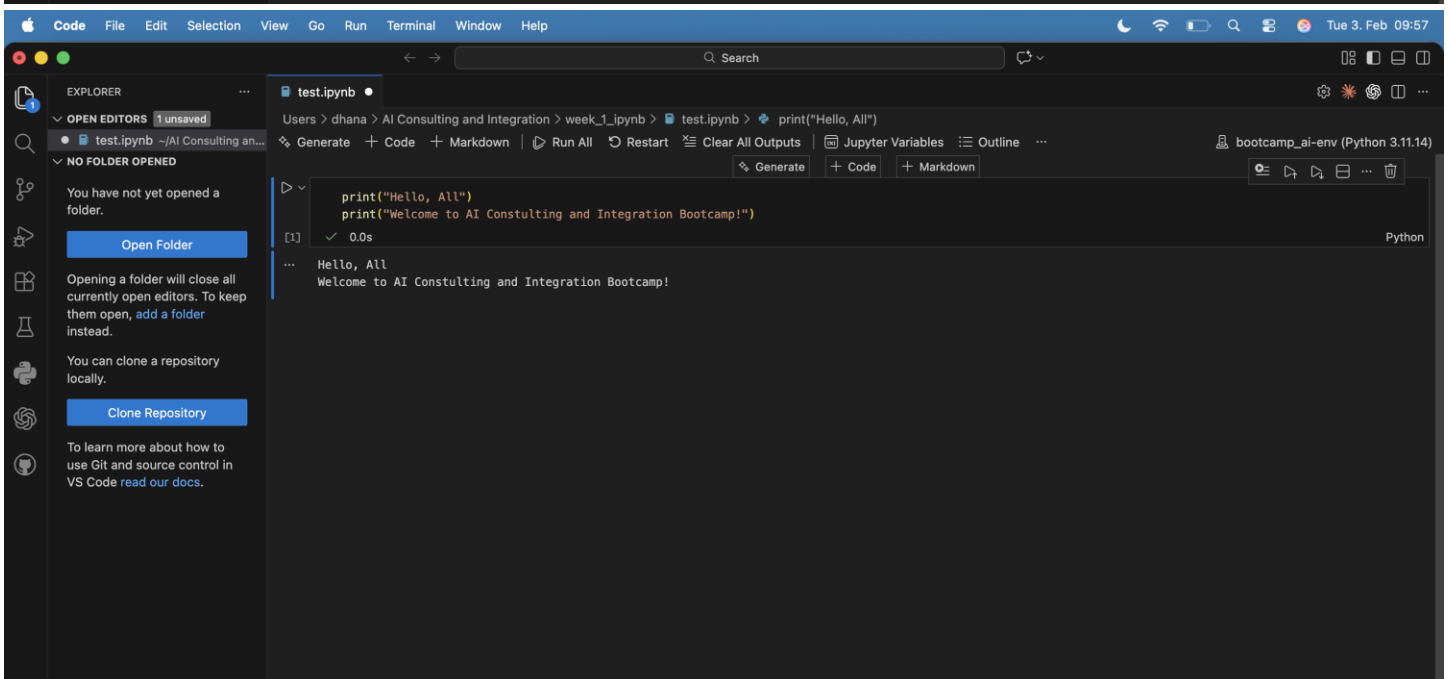
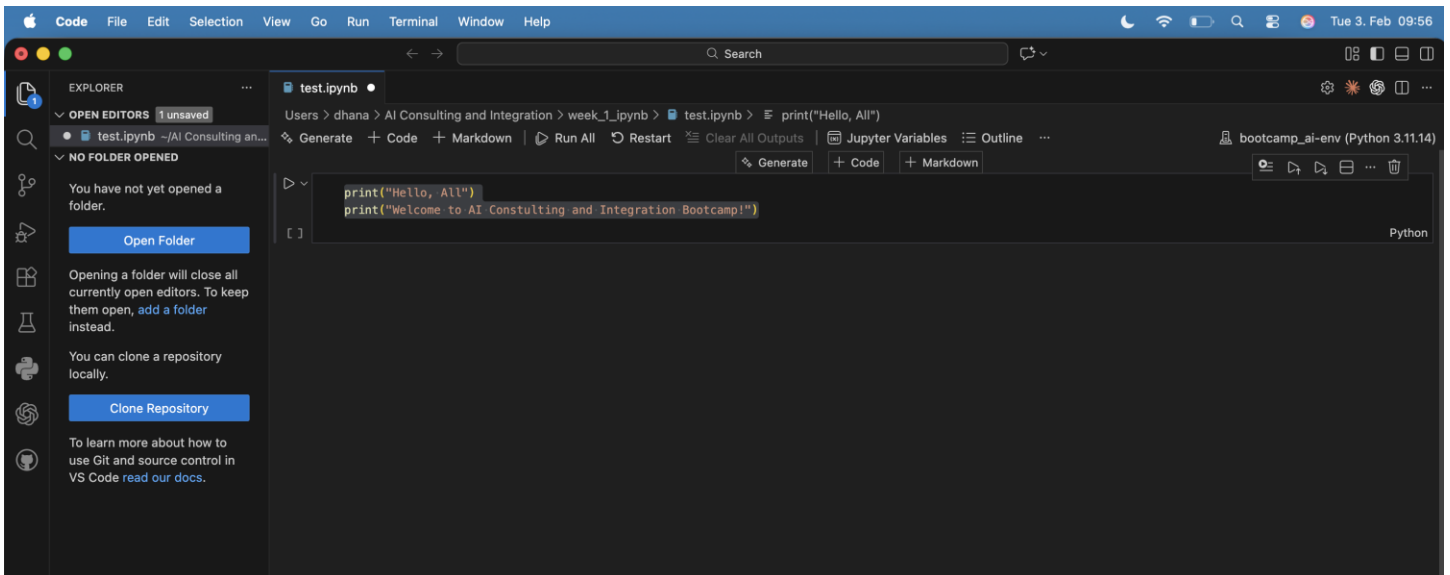
```
print("Hello, All")
```

```
print("Welcome to AI Constulting and Integration Bootcamp!")
```

Select kernel:





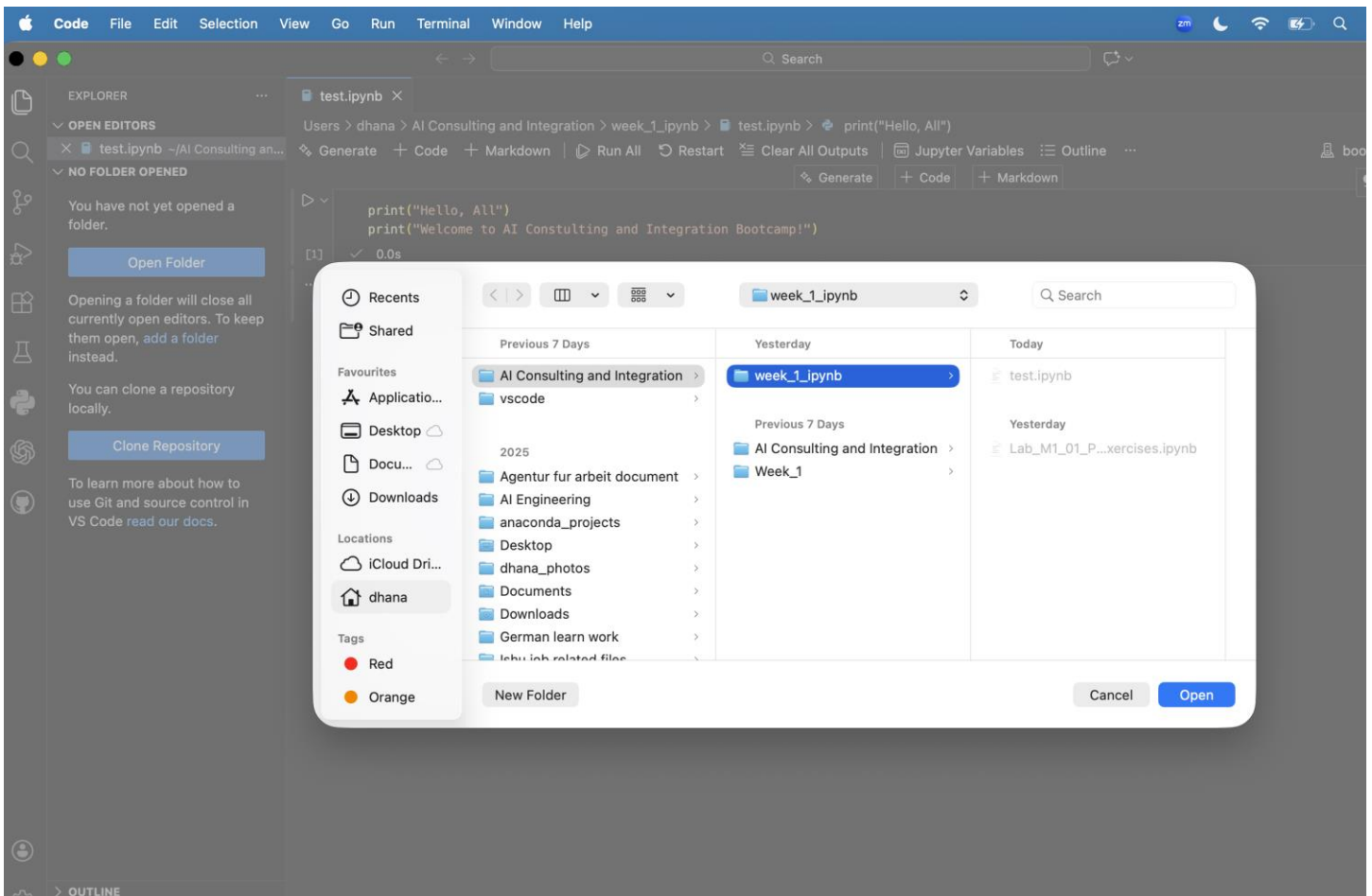
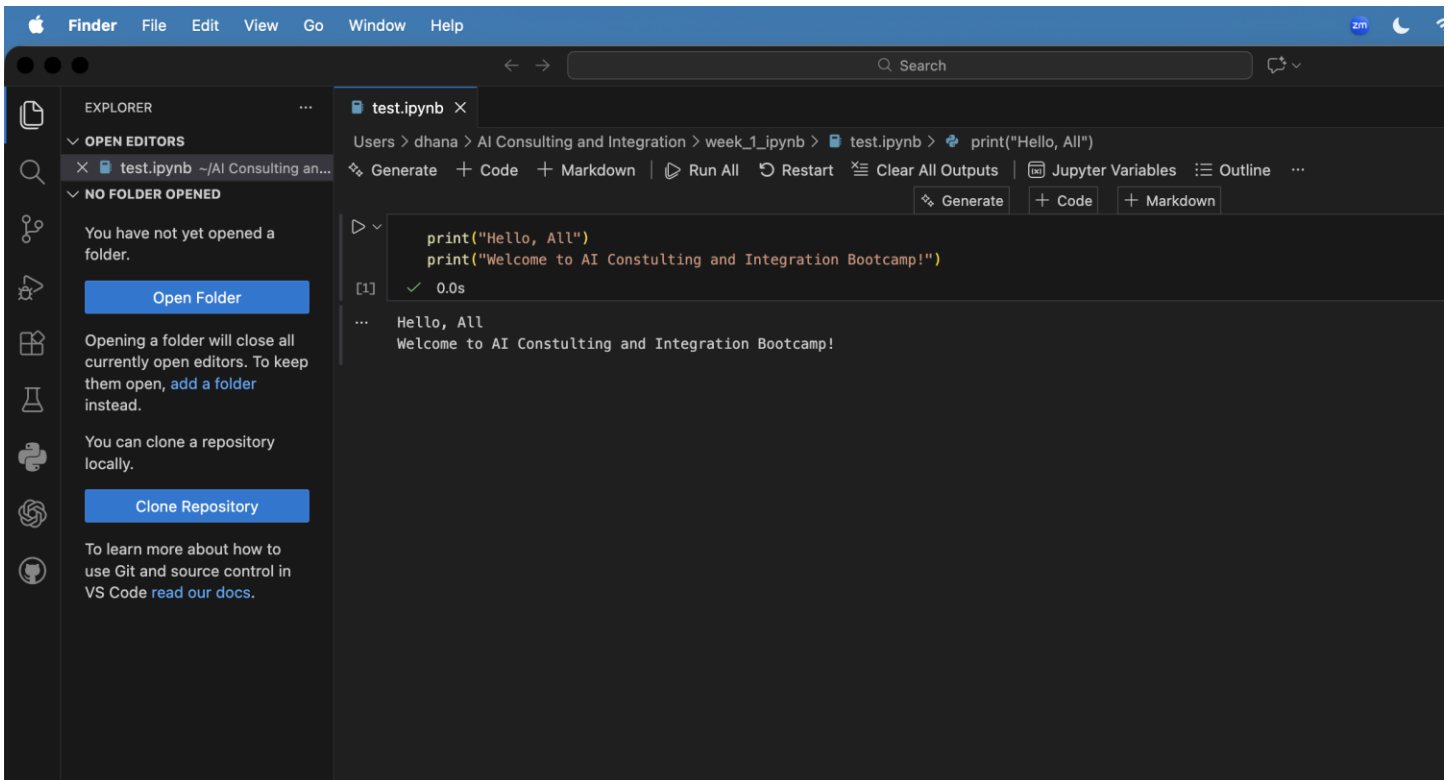


Execute the python code-> Now code got successfully executed.

STEP 10 — First Git Commit (VSCode)

1. Open Folder

- File → Open Folder
- Choose your project folder

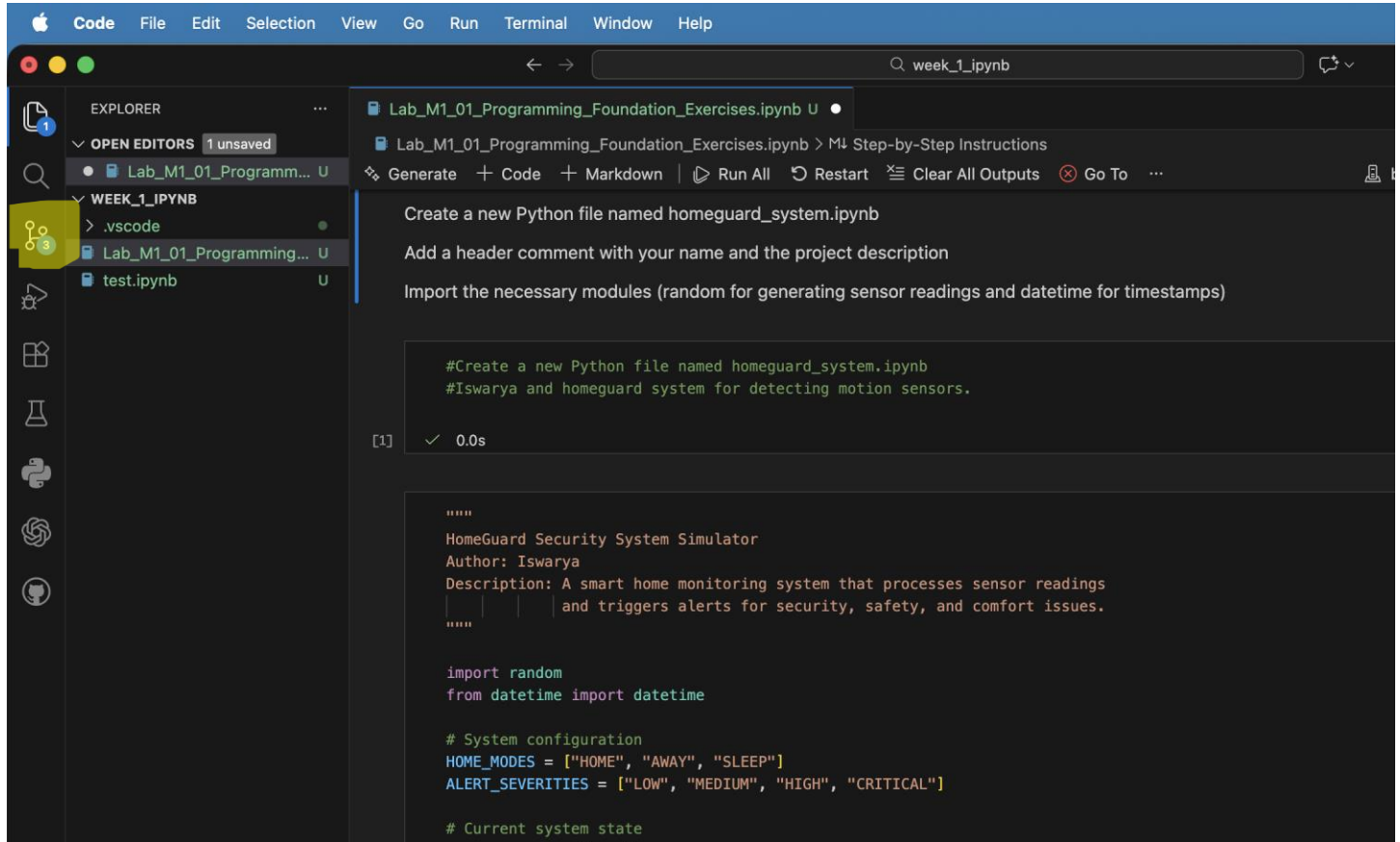


Open the Lab notebook file-> Run all command

2. Initialize Git

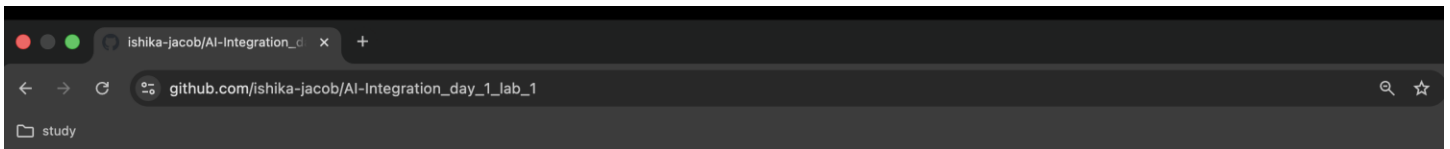
- Click **Source Control** icon
- Click **Initialize Repository**

Left corner -> you will witness source control icon



3. Connect to GitHub

- Source Control ... → **Add Remote**
- Name: origin
- Paste your GitHub repo URL



Set up GitHub Copilot

Use GitHub's AI pair programmer to autocomplete suggestions as you code.

[Get started with GitHub Copilot](#)



Add collaborators to this repository

Search for people using their GitHub username or email address.

[Invite collaborators](#)

Quick setup — if you've done this kind of thing before

[Set up in Desktop](#)

or

[HTTPS](#)

[SSH](#)

https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git



Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# AI-Integration_day_1_lab_1" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git
git push -u origin main
```

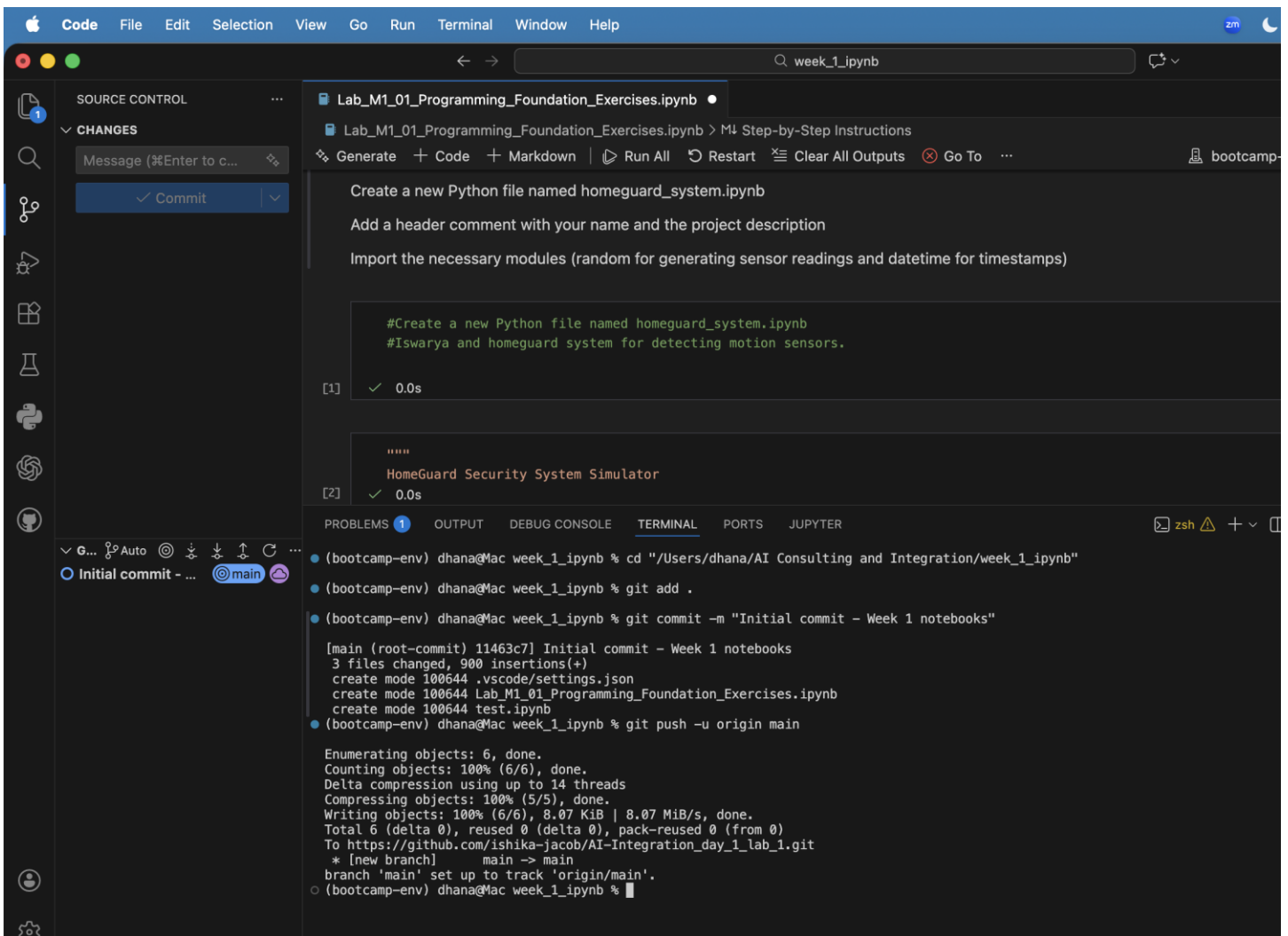


...or push an existing repository from the command line

```
git remote add origin https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git
git branch -M main
git push -u origin main
```



ProTip! Use the URL for this page when adding GitHub as a remote.



4. Commit

- Click + to stage files
- Message: Initial commit
- Press Ctrl + Enter

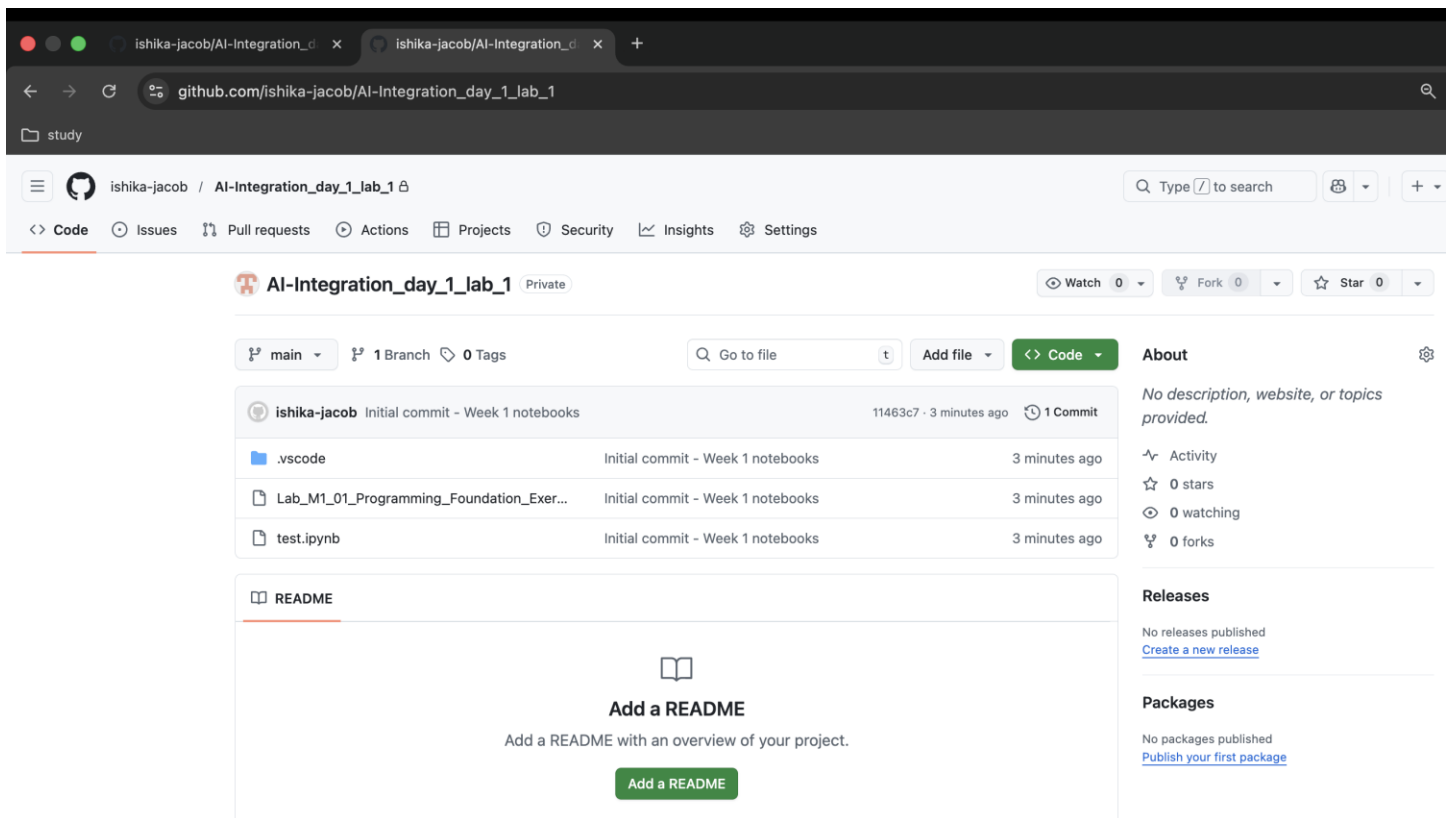
5. Push

- Click **Push**
- Select origin → main

✅ Code is now on GitHub

Open GitHub Account :

https://github.com/ishika-jacob/AI-Integration_day_1_lab_1



Look at the repository page

Once the repo loads:

1. You should see a list of **files and folders**.
2. For your Day 1 lab, you should see:
 - a. Lab_M1_01_Programming_Foundation_Exercises.ipynb
 - b. test.ipynb
 - c. (maybe .gitignore if you added it)
3. If your files are there →  Push worked!

Open VSCode -> Save your notebook

Open VSCode -> save the Notebook Program->

Example code:

```
git init
```

```
git remote add origin https://github.com/ishika-jacob/AI-Integration\_day\_1\_lab\_1.git
```

```
cd "/Users/dhana/AI Consulting and Integration/week_1_ipynb"
```

```
git add .
```

```
git commit -m "Add lab submission comment for PR"
```

git push

Now to check changes in GitHub

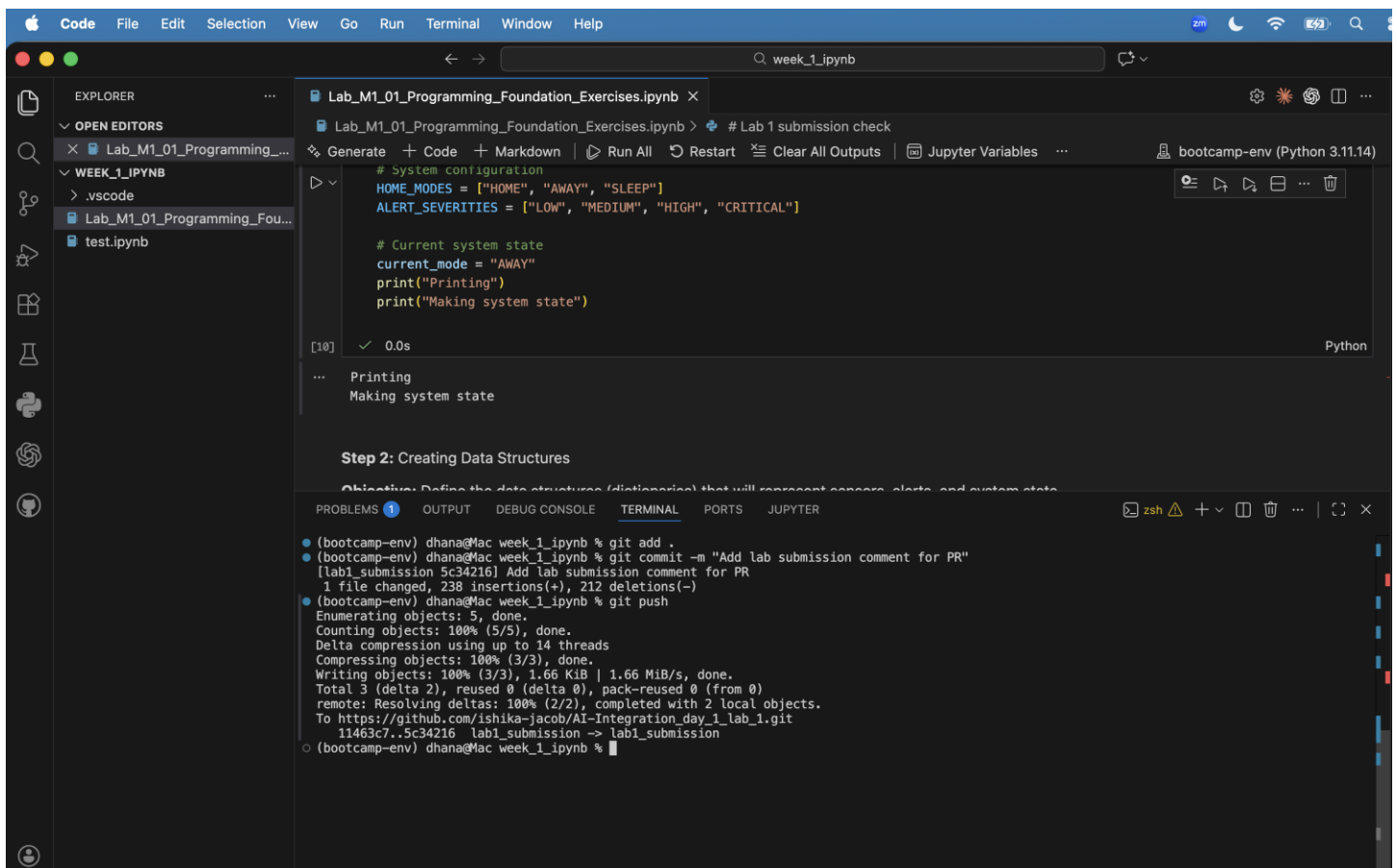
https://github.com/ishika-jacob/AI-Integration_day_1_lab_1

Open Pull request

Or you will be able to see -> compare & pull request

Copy the link from the address bar-> shown in below screenshots

Past the “pull request link from the address bar” -> paste in the “Student portal” to submit the lab



The screenshot shows a VS Code editor window with a Jupyter Notebook open. The notebook is titled "Lab_M1_01_Programming_Foundation_Exercises.ipynb" and contains Python code for a system configuration and state management. The code defines two lists: `HOME_MODES` and `ALERT_SEVERITIES`, and a function to print the current system state. The notebook is running in a "bootcamp-env (Python 3.11.14)" environment. Below the notebook, a terminal window is open, showing the output of a series of git commands: `git add .`, `git commit -m "Add lab submission comment for PR"`, and `git push`. The terminal output shows the progress of the push operation, including object counting, compression, and writing to the remote repository. The final output is a URL for the push operation: `https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git`.

```
# System configuration
HOME_MODES = ["HOME", "AWAY", "SLEEP"]
ALERT_SEVERITIES = ["LOW", "MEDIUM", "HIGH", "CRITICAL"]

# Current system state
current_mode = "AWAY"
print("Printing")
print("Making system state")

[10] ✓ 0.0s Python

... Printing
Making system state

Step 2: Creating Data Structures

Objective: Define the data structures (dictionaries) that will represent sensors, alerts, and system state.

• (bootcamp-env) dhana@Mac week_1_ipynb % git add .
• (bootcamp-env) dhana@Mac week_1_ipynb % git commit -m "Add lab submission comment for PR"
[lab1_submission 5c34216] Add lab submission comment for PR
1 file changed, 238 insertions(+), 212 deletions(-)
• (bootcamp-env) dhana@Mac week_1_ipynb % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 14 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.66 KiB | 1.66 MiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/ishika-jacob/AI-Integration_day_1_lab_1.git
11463c7..5c34216 lab1_submission -> lab1_submission
• (bootcamp-env) dhana@Mac week_1_ipynb %
```

🔒 AI-Integration_day_1_lab_1 Private

👁 Watch 0

🍴 Fork 0

☆ Star 0

🔔 lab1_submission had recent pushes 1 minute ago

Compare & pull request

🔗 main 2 Branches 0 Tags

🔍 Go to file

Add file

<> Code

ishika-jacob	Initial commit - Week 1 notebooks	11463c7 · 18 minutes ago	🕒 1 Commit
📁 .vscode	Initial commit - Week 1 notebooks	18 minutes ago	
📄 Lab_M1_01_Programming_Foundation_Exer...	Initial commit - Week 1 notebooks	18 minutes ago	
📄 test.ipynb	Initial commit - Week 1 notebooks	18 minutes ago	

📖 README

📖

Add a README

Add a README with an overview of your project.

Add a README

About

No description, website, or topics provided.

🔄 Activity

☆ 0 stars

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Releases

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Create a new release

Packages

No packages published

Publish your first package

Languages

Jupyter Notebook 100.0%

ishika-jacob/AI-Integration_day_1_lab_1/pull/1

ishika-jacob / AI-Integration_day_1_lab_1

lab_1_pull_Add lab submission comment for PR #1

Open ishika-jacob wants to merge 1 commit into main from lab1_submission

Conversation 0 Commits 1 Checks 0 Files changed 1 +238 -212

ishika-jacob commented now

No description provided.

Add lab submission comment for PR 5c34216

No conflicts with base branch

Merging can be performed automatically.

Merge pull request You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

Add a comment

Write Preview

Add your comment here...

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications Customize

https://github.com/ishika-jacob/AI-Integration_day_1_lab_1/pull/1

LAB | Programming Foundation Exercises

- ←
2. **Data Structures:** What information did you store in dictionaries, and why was this useful?

3. **Functions:** How did breaking your code into functions make it easier to work with?

4. **Classes:** What advantages did using a class provide over just using dictionaries?

Key takeaways:

- Conditional logic (if/else) is essential for making decisions in programs
- Data structures help organize related information
- Functions make code reusable and easier to maintain
- Classes combine data and behavior into organized units

These concepts form the foundation of all programming and are especially important for AI systems that need to process data and make decisions.

Good luck with your HomeGuard Security System! Remember to test your code frequently and don't hesitate to ask for help if you get stuck.

programming-foundations

Submit the url to your work here:

URL

https://github.com/ishika-jacob/AI-Integration_day_1_lab_1/pull/1

Send lab